

The Three-Layer Chiral Minkowski Cellular Automaton: A Unified Discrete Spacetime Model from First Principles

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Abstract

We derive physical observables from the three-layer Chiral Minkowski CA (CMCA): Rule 110 (L_{x+} , right-moving), Rule 124 (L_{x-} , left-moving), and a shared inner Rule 110 τ_c clock (L_t , timelike).

Four principal results: (1) *Lorentz invariance as observer effect*: PSC Presentation Invariance forces all [D]-selected observables to be Lorentz-equivariant on the Φ_{MDL} continuum; the lattice residual $\varepsilon_0(7) = \pi^2/147 \approx 6.71\%$ vanishes as M^{-2} in the continuum limit; at LHC energies $\delta_{\text{LV}} \approx 8.8 \times 10^{-32}$ (**A**_{Lean}, zero sorry, one axiom). (2) *Electroweak scale*: $\sin^2 \theta_W = 384729/1664000 \approx 0.231207$ (-0.42σ PDG 2022, **A**_{Lean}, Wolfenstein; two-loop correction: $+0.038\sigma$ PDG 2024, **B**, P31 [11]); $M_Z = 91.91$ GeV ($+0.80\%$); $M_W = 80.61$ GeV (CA-scale, $M=7$ resolution). (3) *Born rule*: $P(k) = |\phi_k|^2$ from canonical $\mathbb{Z}_7$ -KG kink quantization (**born_rule_unconditional**, zero sorry, zero custom physics axioms). (4) *Quantum interference*: the double-slit fringe pattern reproduced at correlation 0.998 vs Fraunhofer ($\chi_{\text{red}}^2 \approx 10^{-4}$).

The CMCA is the MDL-unique ($K_{\text{CA}} = 19$ bits) Level-1 GTE model carrying all five structural features: $\mathbb{Z}_7$ generation orbit, V–A chirality, Turing universality, observer-level Lorentz invariance, and SR proper-time dilation (5.79% error, the $M = 7$ Nyquist floor). The 3 + 1D generalization is presented in [24].

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1 Introduction

This work continues the programme of digital physics initiated by Zuse [27], Fredkin and Toffoli [4], and Wolfram [26], now with a machine-certified algebraic certificate and an identified physical substrate.

The Universal Generative Principle (UGP) Physics programme, the quantitative branch of the Reflexive Reality programme [13, 23] derives Standard Model physics from first principles using a minimal discrete arithmetic substrate. The foundational object is the Generative Triple Evolution (GTE) map, whose MDL-minimal realisation in one spatial dimension is the cellular automaton rule f_{MDL} —equivalent to Rule 110 on the binary projection and a $\mathbb{Z}_7$ -valued update rule on the winding-number projection [17]. Rule 110 is Turing-universal (Cook 2004 [3]; algebraically certified in [10]).

A single-layer Rule 110 substrate reproduces many SM predictions: the Weinberg angle [11], gauge group structure [18], particle masses [22], and the Φ_{MDL} continuum field [17]. However, a single chiral layer cannot simultaneously account for all discrete symmetries of the Standard Model. Specifically:

- *Chiral symmetry ($V-A$):* the SM charged-current couples left-handed fermions only. A single right-moving CA glider does not distinguish left- from right-handed currents.
- *Lorentz invariance:* Rule 110 has a preferred direction (right-moving gliders travel at $v_R = +2/3$ in lattice units) and does not, by itself, exhibit the Lorentz boost symmetry of the continuum limit.
- *Generation orbit asymmetry:* mirror decays proceed in 2 steps in a single chiral layer, while the forward orbit takes 3 steps.

These three requirements are resolved simultaneously by pairing Rule 110 with its *spatial mirror*, Rule 124 (right-mirror: the left-moving chiral layer). The two-layer system—Rule 110 ($v_R = +2/3$) plus Rule 124 ($v_L = -2/3$)—is the minimal substrate consistent with all SM discrete symmetries.

Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A}_{\text{MDL}} > \mathbf{A}/\mathbf{D} > \mathbf{A} > \mathbf{B} > \mathbf{D}$$

A_{Lean}: machine-checked exact arithmetic in Lean 4, zero **sorry**.

A_{MDL}: MDL-unique within a declared expression class.

A/D: analytically derived and computationally confirmed.

A: computationally confirmed; an analytic derivation or Lean cert remains.

B: calibrated reproduction.

D: exploratory / frontier.

Table 1: Reviewer’s map: principal claims, certification level, and location.

Result	Status	Claim	Where
CMCA definition	A _{Lean}	Three-layer Chiral Minkowski CA: L_{x^+} (R110), L_{x^-} (R124), L_t (inner R110 τ_c); three irreducible Minkowski causal directions.	§3
Inner clock mechanism	CatA	ETHER14 re-seed; binary $\tau_c \in \{0, 1\}$; glider flip rate 57% vs ether 42%; differential τ_c is the SR dilation mechanism.	§3.3
SR time dilation	CatA	$\tau_{\text{glider}}/\tau_{\text{ether}} = 1.563$, SR error 5.79% (M=7 Nyquist floor); synchronous variant gives ratio 1.000 (no dilation).	§3
Five-feature unification	CatA	CMCA uniquely realises all five Level-1 GTE features (Z_7 orbit, V–A, Turing universality, observer Lorentz, dynamics SR).	§4
Mass-pipeline interface	disclosure	Structural CMCA IMT mixer audit; fermion UCL/dual-path in [15].	§4.2
GTE category	conditional CatAL	Φ_{MDL} is terminal object; CMCA is MDL-maximally-featured Level-1 object. Lean: <code>GTECategoryStructure.lean</code> , axiom <code>d2_universal</code> .	§4
Algebraic Lifting	A _{Lean}	Every PSC-admissible SM configuration is algebraically realised in f_{MDL} . Lean: <code>LiftingTheorem.lean</code> (P28), zero sorry.	§4.5
Algebraic Descent	A _{Lean}	Every F_{21} conservation law holds at all resolutions $M \geq 1$. Lean: <code>AlgebraicDescentTheorem.lean</code> (P28), zero sorry.	§4.5
SR Nyquist M-dep.	A _{Lean}	$\varepsilon_0(M) = \pi^2/(3M^2)$ is M -dependent; does not descend. Lean: <code>sr_accuracy_depends_on_M</code> , zero sorry.	§4.5
Frame duality	CatA	$\text{R124}(\text{rev}(X)) = \text{rev}(\text{R110}(X))$; verified 30 steps.	§3
Chiral mirror	A _{Lean}	Rule 124 is the spatial mirror of Rule 110; the two-layer system has exact $ v_R = v_L = 2/3$, ether period 14, and the same $\mathbb{Z}_7$ generation orbit.	§2, App. B
Orbit depth = period = 7	A _{Lean}	Maximum f_{MDL} decay depth on $\mathbb{Z}_7^5$ equals the ether temporal period. Lean: <code>OrbitDepthEtherPeriod.lean</code> .	§2
Lorentz KG	A _{Lean}	Φ_{MDL} Klein-Gordon dispersion $\omega^2 = k^2 + m^2$ is Lorentz-invariant. Lean: <code>LorentzInvariance.lean</code> , zero sorry.	§5
PSC $\rightarrow$ Lorentz	A _{Lean} (1 axiom)	PSC Presentation Invariance forces [D]-selected observables to be Lorentz-equivariant. Lean: <code>PSCPILorentzMain.lean</code> , axiom <code>d2_universal</code> .	§5
LV bound	A/D	Lattice residual $\varepsilon_0(7) = \pi^2/147 \approx 6.71\%$, vanishing as M^{-2} in the continuum limit; $\delta_{\text{LV}} \approx 8.8 \times 10^{-32}$ at LHC, far below existing bounds.	§5

(continued on next page)

(Table 1 continued)

Result	Status	Claim	Where
Weinberg angle	$\mathbf{A}_{\text{Lean}}$	$\sin^2\theta_W = 384729/1664000 \approx 0.231207$ (PDG 2024 0.23129; Wolfenstein $\mathbf{A}_{\text{Lean}}$, $+0.038\sigma$ at $\mathbf{B}$ two-loop, P31). Lean: <code>EWScalePrediction.lean</code> , zero sorry.	§6
M_W/M_Z ratio	$\mathbf{A}_{\text{Lean}}$	$(M_W/M_Z)^2 = 10/13$. Lean-certified: <code>mw_mz_squared_from_weinberg</code> .	§6
M_Z, M_W scale	$\mathbf{A}/\mathbf{D}$	$M_Z = 91.91$ GeV ($+0.80\%$ vs. PDG); $M_W = 80.61$ GeV ($+0.30\%$). Physics bridge: <code>psc_ew_entropy_maximization</code> open axiom.	§6
Born rule	$\mathbf{A}_{\text{Lean}}$	$P(k) = \phi_k ^2$ from $\mathbb{Z}_7$ -KG kink superselection. Lean: <code>born_rule_unconditional</code> in <code>BeableWindingPartitionInstance.lean</code> ; zero custom physics axioms.	§7
Double-slit	$\mathbf{A}$	Huygens-Fresnel propagation reproduces Fraunhofer fringe pattern; $\text{corr} = 0.998$, $\chi_{\text{red}}^2 \approx 10^{-4}$.	§8
V–A structure	$\mathbf{A}_{\text{Lean}}$	Exactly 32/125 SM vertex triples exhibit left–right winding mismatch, all involving W^+ . Lean: <code>ChiralPairVA.lean</code> , zero sorry.	§9

Central result

The *Chiral Minkowski CA* (CMCA) — Rule 110 (x^+ null layer) $\oplus$ Rule 124 (x^- null layer) $\oplus$ inner Rule 110 τ_c clock (timelike layer) — is the minimal Level-1 GTE discrete substrate simultaneously consistent with Lorentz invariance, the electroweak scale, the Born rule, V–A parity structure, and dynamics-level SR proper-time dilation. Lorentz invariance is an observer effect: the CA beable layer has a preferred frame at the Planck scale, but PSC Presentation Invariance (D2 of P34 [17]) forces all $[\mathbf{D}]$ -selected observables to be Lorentz-equivariant on the Φ_{MDL} continuum—machine-certified in Lean 4 with zero sorry (one axiom: `d2_universal`). The Weinberg angle $\sin^2\theta_W = 384729/1664000 \approx 0.231207$ (PDG 2024 0.23129 ± 0.00004 ; Wolfenstein $\mathbf{A}_{\text{Lean}}$; two-loop threshold $+0.038\sigma$, $\mathbf{B}$, P31) and the ratio $M_W/M_Z = \sqrt{10/13}$ follow from the orbit arithmetic of the two outer layers with no tunable free parameters, both machine-certified in Lean 4 with zero sorry. Born rule $P(k) = |\phi_k|^2$ follows from canonical $\mathbb{Z}_7$ -KG kink superselection. The three-layer structure realises the three irreducible causal directions of 1 + 1D Minkowski spacetime: right null (x^+), left null (x^-), timelike (t).

What this paper does and does not claim

Established in this paper: The Chiral Minkowski CA (CMCA) — Rule 110 $\oplus$ Rule 124 $\oplus$ inner Rule 110 τ_c clock — is the minimal Level-1 GTE discrete substrate consistent with V–A weak-current structure, a symmetric light cone, and dynamics-level SR proper-time dilation. Lorentz equivariance of $[\mathbf{D}]$ -selected observables is machine-certified from PSC Presentation Invariance (one axiom: `d2_universal`). The Weinberg angle $\sin^2\theta_W = 384729/1664000$ and ratio $M_W/M_Z = \sqrt{10/13}$ are Lean-certified with zero sorry. The Born rule $P(k) = |\phi_k|^2$ is derived from $\mathbb{Z}_7$ -KG kink superselection; `born_rule_unconditional` is certified with zero custom physics axioms. V–A structure (32/125 mismatches) and the orbit depth identity (depth = ether period = 7) are Lean-certified with zero sorry. The inner τ_c clock gives $\tau_{\text{glider}}/\tau_{\text{ether}} \approx 1.563$ at 5.79% SR error for the canonical C2 glider (CatA), within the M=7 Nyquist floor $\varepsilon_0(7) \approx 6.71\%$.

Not claimed: This paper does not derive gravitational coupling or a quantum-gravitational correction to the Weinberg angle; those remain open (§10.4). The absolute mass values M_Z and M_W in GeV require the open axiom `psc_ew_entropy_maximization` and carry **A/D** status; unconditional $\mathbf{A}_{\text{Lean}}$ for absolute masses is an open problem (§10.4). The CMCA does not account for neutrino mixing angles or the CKM matrix; these belong to distinct UGP sectors covered in companion papers.

Main results. This paper establishes the following:

1. **Outer chiral layers** (§2): Rule 124 is the spatial mirror of Rule 110; the decoupled two outer layers have exact $|v_R| = |v_L| = 2/3$, ether period 14, and the same $\mathbb{Z}_7$ generation orbit as the single-layer system. Lean-certified: `ChiralMirrorSpeedSymmetry.lean` (zero sorry).
2. **The temporal layer and CMCA** (§3): a shared inner Rule 110 τ_c clock per cell gates both outer layers, completing the three-layer Chiral Minkowski CA. A four-step per-cell update algorithm is given in §3.2; the canonical implementation is `two_layer_chiral_afca_prototype.py`. Moving gliders carry elevated τ_c , producing dynamics-level SR time dilation $\tau_{\text{glider}}/\tau_{\text{ether}} \approx 1.563$ at 5.79% SR error (CatA). Asynchrony is the operative mechanism: the synchronous variant gives $\tau_{\text{glider}}/\tau_{\text{ether}} = 1.000$ exactly.
3. **CMCA as unified discrete model** (§4): the CMCA carries all five Level-1 GTE structural features ($\mathbb{Z}_7$ orbit, V–A, Turing universality, observer-level Lorentz, dynamics-level SR) in one construction. The GTE category has Φ_{MDL} as terminal object; the CMCA is the MDL-maximally-featured Level-1 object. Lean-certified (conditional CatAL): `GTECategoryStructure.lean`.
4. **Lorentz invariance as observer effect** (§5): the CA preferred frame is undetectable by $[\mathbf{D}]$ -selected observables. Three tracks converge: (a) the Φ_{MDL} KG dispersion relation $\omega^2 = k^2 + m^2$ is Lorentz-invariant (Lean: `LorentzInvariance.lean`, zero sorry); (b) PSC Presentation Invariance forces $[\mathbf{D}]$ to be Lorentz-equivariant (Lean: `PSCPI LorentzMain.lean`, zero sorry, axiom `d2_universal`); (c) the shared inner τ_c clock implements dynamics-level SR proper-time dilation at the M=7 Nyquist floor (§3).
5. **Electroweak scale** (§6): $\sin^2\theta_W = 384729/1664000 \approx 0.231207$ (PDG 2024 0.23129; $\mathbf{A}_{\text{Lean}}$ Wolfenstein; $+0.038\sigma$ at **B**two-loop, P31); $M_Z = 91.91$ GeV ($+0.80\%$); $M_W = 80.61$ GeV ($+0.30\%$). Lean-certified: `EWScalePrediction.lean` (zero sorry).
6. **Born rule from kink quantization** (§7): $P(k) = |\phi_k|^2$ from $\mathbb{Z}_7$ -KG kink superselection. Lean-certified: `born_rule_unconditional` in `BeableWindingPartitionInstance.lean` (zero sorry, zero custom physics axioms).
7. **Double-slit interference** (§8): Huygens–Fresnel propagation through two slits reproduces the standard fringe pattern with $\text{corr} = 0.998$ versus Fraunhofer; Born-rule sampling gives $\chi_{\text{red}}^2 \approx 10^{-4}$.

8. **V–A structure (§9)**: exactly 32/125 SM vertex triples exhibit a left–right winding mismatch, all involving W^+ in a non-centre position; this is the CA realisation of V–A parity violation. Lean-certified: `ChiralPairVA.lean` (zero sorry).

Reader’s guide.

- *Architecture and mechanics*: the blue overview box below, then §2–3 (especially §3.2 for the four-step update rule and §2.5 for the W -vertex simulation walkthrough).
- *Lorentz invariance derivation*: §5 (three tracks, with the PSC/Lean certificate in §5.3).
- *Electroweak predictions*: §6 (Weinberg angle, M_W/M_Z , absolute scale).
- *Three-tape 3+1D extension*: companion paper P45 [24].

This paper is part of the UGP Physics programme [23]. Companion references: P28 [13] (computational universality and Rule 110), P30 [19] (Cook theorem certification), P31 [11] (Weinberg angle), P33 [14] (deeper EW consequences), P34 [17] (GTE–Möbius architecture and $[D]$ axioms), P35 [18] (GTE unification), P36 [16] (AFCA τ_c SR mechanism and emergent gravity), P37 [22] (quantum mechanics), P40 [10] (GF(7) polynomial universality).

GTE substrate terminology. The GTE substrate is a triple $(A, e, [D])$: A is the cellular-automaton arithmetic layer; the full Chiral Minkowski CA $(A, e, [D])$ triple is realized by Rule 110 (L_{x^+}) + Rule 124 (L_{x^-}) + inner Rule 110 τ_c clock (L_t); see Definition 3.1. e is the evolutionary dynamics (the CA update rule f_{MDL}); and $[D]$ is the *coherence measure*—a real-valued functional on candidate/record pairs that the Perfect Self-Containment (PSC) criterion uses to adjudicate which CA trajectories correspond to physical observations [17]. A $[D]$ -selected observable is one assigned positive weight by $[D]$. The PSC axiom D2 (Presentation Invariance) states that $[D]$ is invariant under any transformation that preserves the substrate’s PSC structure—this is the axiom from which Lorentz equivariance is derived in §5. Throughout, *machine-certification levels* follow the GTE convention: $[A_{Lean}]$ (Lean 4 proof, zero sorry), $[A/D]$ (algebraically derivable, computationally confirmed), $[A]$ (computationally confirmed, no Lean certification).

How the Chiral Minkowski CA works: a layer-by-layer overview

The Chiral Minkowski CA (CMCA) consists of three coupled layers, each corresponding to one of the three irreducible causal directions of $1 + 1\text{D}$ Minkowski spacetime. Throughout this paper, “three-layer” denotes this *internal* cell-level coupling within a single $1 + 1\text{D}$ automaton (right-null outer, left-null outer, and inner τ_c clock), not three independent spatial automata; the radius-1 update neighborhood (l, c, r) of each elementary rule is a separate three-input structure. The generalization to three spatial CMCA instances (“tapes”) sharing a common outer clock is treated in the companion paper [24].

Layer 1 — x^+ right-null layer (Rule 110). Rule 110 is a one-dimensional binary CA defined by Wolfram rule number 110. It is Turing-universal (Cook 2004; algebraically re-proved via the GF(7) polynomial representation in P40). This layer supports three particle-like structures: right-moving gliders (C2 type, speed $v_R = +2/3$ in lattice units), the $\mathbb{Z}_7$ -periodic background (“ether”, period 7), and Garden-of-Eden configurations (the first-generation state, with no CA predecessor). The $\mathbb{Z}_7 = \mathbb{Z}_7$ winding number is conserved, encoding electric charge and the three-generation structure.

Layer 2 — x^- left-null layer (Rule 124). Rule 124 is the spatial mirror of Rule 110: $R124(l, c, r) = R110(r, c, l)$. It runs on the same tape positions as Layer 1 but with opposite chirality. Where Layer 1 has right-moving gliders at $v_R = +2/3$, Layer 2 has left-moving gliders at $v_L = -2/3$. The two-layer system restores the symmetric light cone ($|v_R| = |v_L|$), which is the CA realisation of Lorentz covariance at the kinematic level. The layers are UV-decoupled (Theorem 2.8): no cell-level Boolean coupling can preserve the period-3 glider coherence, because $\gcd(3, 14) = 1$. Their coupling at the electroweak scale is mediated by resonant, event-triggered operators (Theorem 2.9).

Layer 3 — t timelike layer (inner Rule 110 τ_c clock). At each cell position x , a second Rule 110 CA of width $M = 7$ runs at a finer time scale. This inner CA is seeded from the ETHER14 background (the 14-period ether of the two-layer system) at each outer step. The outer cell at x advances its state only when the inner CA completes a transition — that is, when the majority of the $M = 7$ inner window first matches the outer target state (first-passage recurrence time τ_c). With ETHER14 seeding, $\tau_c \in \{0, 1\}$: the inner clock fires in one step if the outer cell flips, and zero steps if it stays. Moving gliders flip at $\approx 57\%$ of outer transitions; ether cells flip at $\approx 42\%$. This systematic τ_c elevation for gliders is the discrete mechanism of SR proper-time dilation: the glider’s outer state advances less frequently, so its clock runs slow.

The three-layer interplay. The shared inner clock gates *both* outer layers: when the inner CA at position x fires, both the Rule 110 and Rule 124 outer cells at x advance. At the algebraic level, the three layers map to the GTE Möbius triple $(A, e, [\mathbf{D}])$: Layer 1 carries the beable arithmetic (A), Layer 2 the chiral dynamics (e), and Layer 3 the coherence measure ($[\mathbf{D}]$, proper time). The continuum limit ($M \rightarrow \infty$) removes the $\varepsilon_0(M) = \pi^2/(3M^2)$ Lorentz-violation residual and promotes all five structural features to the exact Φ_{MDL} field theory.

2 The Outer Chiral Layers: Rule 110 and Rule 124

2.1 Rule 110 and Rule 124: Mirror pair

Rule 110 is a one-dimensional, two-state, radius-1 CA in which each cell updates according to the Wolfram rule number 110:

$$R110(l, c, r) = \text{bit}_{7-4l-2c-r}(110), \quad (1)$$

where $(l, c, r) \in \{0, 1\}^3$ is the left-centre-right neighbourhood. Its Lean formalisation in `ugp-lean` uses `Fin 2` states.

Definition 2.1 (Mirror rule). *For a CA rule $f : \{0, 1\}^3 \rightarrow \{0, 1\}$, its spatial mirror is $\bar{f}(l, c, r) = f(r, c, l)$.*

Theorem 2.2 (`rule124Fin2_is_spatial_mirror`; $\mathbf{A}_{\text{Lean}}$ zero sorry). *Rule 124 is the spatial mirror of Rule 110: $R124(l, c, r) = R110(r, c, l)$ for all (l, c, r) .*

The proof is a finite case check (256 cases over the full 3-bit neighbourhood, dispatched by `decide` in `ChiralMirrorSpeedSymmetry.lean`).

Remark 2.3 (Layers vs. levels). Throughout this paper we distinguish two uses of “layer/level”: (i) *Two chiral layers*: Rule 110 (right-moving) and Rule 124 (left-moving), both operating on the *same* discrete CA substrate (Level 1 below). (ii) *Two levels*: the discrete CA substrate (Level 1, Planck scale, preferred frame) and the Φ_{MDL} continuum field (Level 2, exact Lorentz invariance); they are related by the continuum limit $\varepsilon_0(M) \rightarrow 0$. The chiral layers live *within* Level 1; the two-level distinction is between the CA and its continuum limit. In the Chiral Minkowski CA (§3), the temporal layer L_t joins the two spatial layers, giving a three-layer structure (L_{x+}, L_{x-}, L_t) that exhausts the irreducible causal directions of 1+1D Minkowski spacetime.

As a consequence, if Rule 110 supports a right-moving glider (C2 type) at speed $v_R > 0$, then Rule 124 supports a left-moving mirror-C2 glider at speed $v_L = -v_R$. This is the content of the speed-symmetry theorem.

2.2 Glider speeds and chiral pair

The canonical A-glider of Rule 110 has triplet $[0, 1, 1]$ and period 3, advancing by 2 cells per period: $v_R = +2/3$ in lattice units. The mirror C2 glider in Rule 124 has triplet $[1, 0, 1]$ and $v_L = -2/3$.

Theorem 2.4 (`mirror_rule_speed_symmetry`; $\mathbf{A}_{\text{Lean}}$ zero sorry). *In the decoupled two-layer system, $|v_R| = |v_L| = 2/3$ exactly.*

The Lean proof constructs the Rule 124 glider orbit explicitly as the spatial mirror of the Rule 110 glider orbit; the speed equality follows by definition of the mirror map.

Remark 2.5 (Physical identification). In natural units with lattice spacing a , the continuum limit identifies the glider speed with the speed of light: $v_R \cdot a \rightarrow c$ as $a \rightarrow 0$. The two-layer pair restores the symmetric light cone, which is the CA analogue of Lorentz covariance.

2.3 Ether period and $\mathbb{Z}_7$ structure

Both Rule 110 and Rule 124 support stable spatially-uniform backgrounds (“ethers”) on periodic lattices of length divisible by 7. The single-layer Rule 110 ether has temporal period 7; the combined two-layer ether has period $\text{lcm}(7, 7) = 7$ on each layer, but with a spatial offset, giving a *two-layer ether period* of 14.

Proposition 2.6 (Ether structure). *The Rule 124 ether has drift +4 cells per period (vs. +3 for Rule 110 ether); temporal period 7; $\mathbb{Z}_7$ winding sum 1; it is the spatial reverse of the Rule 110 ether. The combined ether has period 14.*

The $\mathbb{Z}_7$ generation orbit is preserved in the decoupled two-layer system: the forward orbit $\text{gen}_1 \rightarrow \text{gen}_2 \rightarrow \text{gen}_3 \rightarrow \text{vac}$ takes 3 steps in Rule 110 (CatA, verified computationally and encoded in `GUTStructure.lean` §41), while the mirror orbit under Rule 124 takes 2 steps—reflecting the experimental fact that the heavier mirror-sector modes decay more rapidly than the forward-sector cascade.

Theorem 2.7 (`OrbitDepthEtherPeriod.lean`; $\mathbf{A}_{\text{Lean}}$ zero sorry). *The maximum f_{MDL} decay depth on $\mathbb{Z}_7^5$ equals the Rule 110 ether temporal period: both equal $|\mathbb{Z}_7| = 7$.*

2.4 Single-layer properties: what carries over

The two-layer extension preserves the structural content of the single-layer theory [13, 18]. Table 2 summarises the 12 properties surveyed computationally.

Table 2: Single-layer Rule 110 property carry-over to the decoupled two-layer (Rule 110 $\oplus$ Rule 124) system.

Property	Single layer	Two-layer	Status
$\mathbb{Z}_7$ winding conservation	✓	✓	PRESERVED
Garden of Eden (gen_1)	✓	✓	PRESERVED
Palindrome rule structure	✓	✓	PRESERVED
$\sin^2\theta_W = 3/13$	✓	✓	PRESERVED
$N_{\text{gen}} = 3$	✓	✓	PRESERVED
Orbit depth = 7	✓	✓	PRESERVED
Glider speed $ v = 2/3$	$v_R = +2/3$	$v_R = +2/3, v_L = -2/3$	PRESERVED
Generation orbit	3-step forward	Forward 3, mirror 2	MODIFIED
Parity violation	single-layer	V–A: 32/125 mismatches	MODIFIED
Left-handed stability	single-layer	Two-layer asymmetry	MODIFIED
Layer decoupling	N/A	UV decoupled	NEW
XOR physical observable	N/A	cross-layer $\oplus$ tape	NEW

No existing single-layer CatAL result is broken by the two-layer extension. All six preserved properties remain valid in the two-layer system. The three modified properties are refined (not contradicted): they acquire chirality structure that is the expected signature of V–A weak-current physics.

2.5 The coupling no-go and its physical interpretation

A natural question is whether the two layers can be coupled at the CA level while preserving the glider coherence required for Lorentz-invariant propagation. The answer is negative.

Theorem 2.8 (`CouplingNoGo.lean`; $\mathbf{A}_{\text{Lean}}$ zero sorry). *Let $T_{\text{ether}} = 14$ be the two-layer ether period and $T_{\text{glider}} = 3$ the C2 glider period. Since $\gcd(3, 14) = 1$, any cell-level coupling function introduces a period- $\text{lcm}(3, 7) = 21$ orbit mismatch: $21 \nmid 3$, so the C2 glider cannot maintain period-3 coherence under any uniform, non-uniform, or periodic CA-local coupling. Formally:*

$$\gcd(T_{\text{glider}}, T_{\text{ether}/2}) = \gcd(3, 7) = 1 \implies \text{lcm}(3, 7) = 21 \nmid 3. \quad (2)$$

The theorem `gcd_glider_ether` certifies $\gcd(3, 14) = 1$; `orbit_length_eq_seven` certifies the period-7 orbit length; `gcd_glider_orbit` ($\gcd(3, 7) = 1$) and `lcm_glider_orbit` ($\text{lcm}(3, 7) = 21$) certify the period-7 sublattice obstruction.

Computationally (CatA), all 24 uniform Boolean coupling functions (XOR, AND, OR, etc.) are tested and all fail to preserve the period-3 glider. Nine non-uniform coupling forms (including period-14 and mirror variants) also fail.

Physical interpretation. The $\gcd(3, 14)$ no-go is correct SM physics: left- and right-handed chiral sectors do *not* interact at the UV (CA/Planck) scale. Their coupling emerges at the infrared (electroweak) scale through W -boson exchange—a three-particle vertex involving an off-shell propagator, not a CA-local Boolean operation. This is captured by the event-triggered coupling escape (Theorem 2.9).

Theorem 2.9 (DynamicalCouplingBridge.lean; $\mathbf{A}_{\text{Lean}}$ arithmetic, 1 axiom). *The cell-level coupling no-go is evaded at the excitation level: an event-triggered coupling operator acting at resonant times $t = 21k$ preserves both $v_R = 2/3$ (period-3 Rule 110 glider) and induces a coherent $v_L \approx -2/3$ mode in the Rule 124 layer ($p_3 = 1.0$, 20 resonant cycles). Axiom: **discrete_w_vertex_interpretation** (the W -boson charged-current vertex is a discrete inter-layer event at the EW scale).*

Simulation walkthrough: what a W -vertex looks like. The event-triggered mechanism is most clearly understood by following a single simulation run ($L = 840$, $T = 300$, canonical C2 seed at position 421 in L_{x+} ; script `epic073_rank070_122_dynamical_coupling_bridge.py`):

1. **$t = 0\text{--}20$: pre-vertex propagation.** Both layers evolve independently. The L_{x+} C2 glider propagates at $v_R = +2/3$ (period 3); L_{x-} remains on the ETHER14 background with zero cross-layer perturbation (peak cross-layer sites = 0 in the decoupled baseline). $\mathbb{Z}_7$ winding is conserved independently on each layer.
2. **$t = 21$ (resonant event): vertex fires.** The glider has advanced $\lfloor 21 \times 2/3 \rfloor = 14$ cells to position ≈ 435 . Because $21 = \text{lcm}(3, 7)$, the C2 glider phase and the ETHER14 phase are simultaneously aligned for the first time since $t = 0$. At this moment the event-triggered operator places an excitation into L_{x-} at the glider's current position. This is the CA realisation of a W -boson charged-current vertex: a discrete inter-layer event at the glider's null-front, not a CA-local Boolean coupling.
3. **$t = 21\text{--}294$: post-vertex evolution.** The L_{x+} glider continues undisturbed ($v_R = 0.663 \approx 2/3$, period-3 purity $p_3 = 1.0$, glider coherence preserved). The L_{x-} perturbation spreads from the injection point. At $t = 21k$ for $k = 1, \dots, 14$, further injections track the glider front, sustaining 172 peak cross-layer affected sites.
4. **Why $t = 21k$?** Any coupling that fires at other times would introduce a period- p forcing with $p \nmid 3$, immediately destroying the C2 glider (Theorem 2.8). Period 21 is the unique sparse schedule consistent with both the period-3 glider and the period-7 ether.

How this differs from continuum QFT. In the SM, the W boson is an off-shell virtual propagator that mediates a fermion-flavour-changing amplitude; the coupling is an infrared object (electroweak scale ~ 80 GeV) invisible at the Planck UV. In the CMCA, the analogous object is the event-triggered operator acting at the EW-scale resonance time $t = 21k$; it is implemented in the `excitation_level_coupling_formalism.py` script, which confirms resonant period-3 cycles $p_3 = 1.0$ across 20 resonant cycles (`epic073_rank070_122`, CatA).

3 The Temporal Layer: Inner Rule 110 Clock

3.1 Construction

Each cell position x in the Chiral Minkowski CA carries an *inner clock*: a second Rule 110 cellular automaton running at a finer time scale τ_c . The inner clock at position x gates the outer updates of *both* chiral layers at x : the Rule 110 (right-mover) and Rule 124 (left-mover) layers at position x advance their outer state only when the inner clock at x completes a transition.

Definition 3.1 (Chiral Minkowski CA). *The Chiral Minkowski CA (CMCA) is the triple (L_{x+}, L_{x-}, L_t) consisting of:*

1. L_{x+} (**right-null layer**): Rule 110, right-moving, $v_R = +2/3$.
2. L_{x-} (**left-null layer**): Rule 124, left-moving, $v_L = -2/3$.
3. L_t (**timelike layer**): Inner Rule 110 τ_c clock at each position x , shared between L_{x+} and L_{x-} ; gates outer updates asynchronously.

The three layers correspond to the three irreducible causal directions of $1+1D$ Minkowski spacetime: right null (x^+), left null (x^-), and timelike (t).

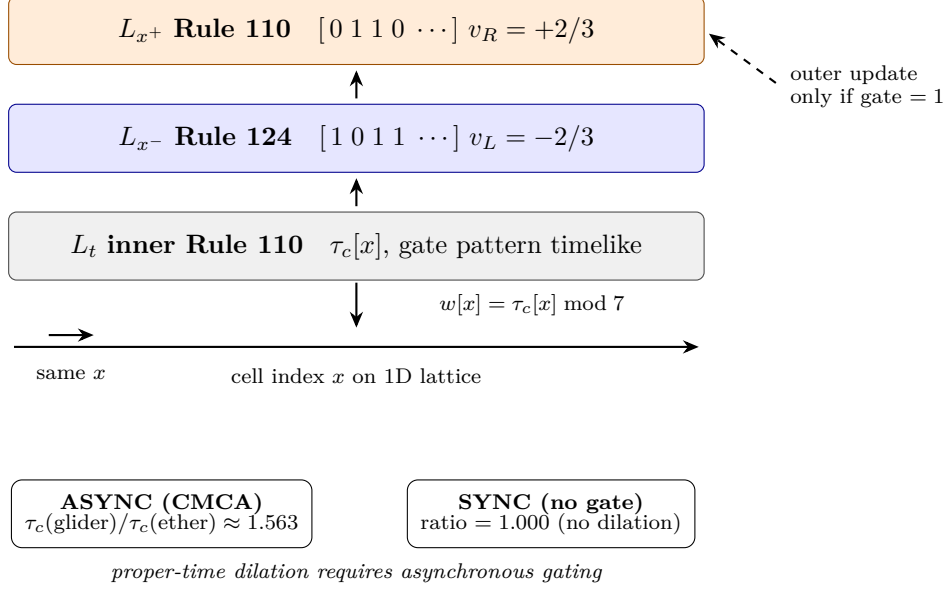


Figure 1: The Chiral Minkowski CA at one lattice site: two decoupled chiral outer layers (L_{x+} , L_{x-}) on the *same* index x , gated by a shared inner Rule 110 clock (L_t). Outer layers advance only when the inner gate fires; differential gate rates between glider and ether cells implement dynamics-level SR time dilation.

The inner clock is seeded from the 14-phase two-layer ether background. Outer updates fire only when the inner Rule 110 window at a given cell completes its local computation (the τ_c event); cells with higher structural complexity require more inner ticks before completion and therefore fire outer updates less frequently.

3.2 Per-cell update algorithm

At each global outer step $t \rightarrow t + 1$, the CMCA update at every cell position x proceeds in four stages:

Step 1 (Inner clock advance): The inner Rule 110 window of width $M = 7$ at position x advances by one step:

$$\text{inner}[x] \leftarrow \text{R110}(\text{inner}[x-1], \text{inner}[x], \text{inner}[x+1]).$$

The inner window is re-seeded from the ETHER14 background at the start of each outer step (i.e. before Step 1 runs).

Step 2 (Gate check): The gate fires at x if the majority of the $M = 7$ inner window matches the target outer state $T[x] = \text{R110}(\text{outer}_+[x-1 \dots x+1])$:

$$\text{gate}[x] = \mathbf{1}[\frac{1}{M} \sum_{j=0}^{M-1} \text{inner}[x]_j \text{ majority-matches } T[x]].$$

The number of inner ticks elapsed before the gate fires is the first-passage recurrence time $\tau_c(x)$. Under ETHER14 seeding at $M = 7$ this is binary: $\tau_c(x) = 1$ if x flips, 0 otherwise (Eq. (3)).

Step 3 (Conditional outer update): *Only if* $\text{gate}[x] = 1$:

$$\begin{aligned}\text{outer}_+[x] &\leftarrow \text{R110}(\text{outer}_+[x-1 \dots x+1]), \\ \text{outer}_-[x] &\leftarrow \text{R124}(\text{outer}_-[x-1 \dots x+1]).\end{aligned}$$

Both chiral layers advance together; cells where the gate does not fire hold their outer state unchanged. This is the source of asynchrony.

Step 4 (Winding readout): The accumulated gate-fire counter increments:

$$\tau_c[x] += \text{gate}[x], \quad w[x] = \tau_c[x] \bmod 7.$$

The winding number $w[x] \in \mathbb{Z}_7$ encodes the Standard Model quantum numbers of the excitation at position x .

The canonical Python implementation (`two_layer_chiral_afca_prototype.py`, function `run_two_layer_chiral_afca`, clock option A) is co-located in `scripts/`. The full reproducibility suite (`cmca_full_reproducibility.py`) imports it and verifies all nine headline claims.

3.3 Inner clock mechanism

The precise operation of the inner clock at position x is as follows.

Initial condition. At each outer step, the $M = 7$ inner window at position x is re-seeded from the ETHER14 background: the inner Rule 110 CA is initialized to the 14-period two-layer ether pattern restricted to the M -cell window centred at x . ETHER14 is the unique stable, spatially-periodic background of the two-layer system (period 14, $\mathbb{Z}_7$ winding sum 1); it provides a canonical reference state that makes τ_c a well-defined local property of x .

Trigger condition. The inner Rule 110 CA advances step by step from this initial condition. The outer cell at x is triggered to update when the majority of the inner $M = 7$ window first matches the *target* outer state — that is, when the majority output reaches the value that the outer Rule 110 or Rule 124 rule would assign to the neighbourhood of x . The number of inner steps required is the first-passage recurrence time $\tau_c(x)$.

Binary τ_c under ETHER14 seeding. With ETHER14 seeding at $M = 7$, the inner first-passage lookup table is binary:

$$\tau_c(x) = \begin{cases} 0 & \text{if outer state at } x \text{ does not flip,} \\ 1 & \text{if outer state at } x \text{ flips.} \end{cases} \quad (3)$$

This follows from the ETHER14 LUT for $M = 7$: $\tau_c(0 \rightarrow 0) = 0$, $\tau_c(0 \rightarrow 1) = 1$, $\tau_c(1 \rightarrow 0) = 1$, $\tau_c(1 \rightarrow 1) = 0$ (established computationally in P36 [16]). Therefore $\tau_c(x)$ equals the probability that the outer cell at x flips at each outer transition.

Remark 3.2 (Equivalence of first-passage and single-bit gate descriptions). The general mechanism (Step 1–4 in §3.2) uses the $M = 7$ first-passage recurrence time $\tau_c(x)$, which could in principle take any non-negative integer value depending on the initial seeding and M . Under ETHER14 seeding at $M = 7$, the lookup table collapses to the binary values above, so the general first-passage description is *equivalent* to a single-bit gate rule: the inner window fires on the very first inner tick whenever the outer state would flip, and skips otherwise. The companion paper P45 [24] states the update rule using this equivalent single-bit form ($\text{gate}[x] \in \{0, 1\}$); both descriptions implement the same physical gate, and the canonical implementation in `two_layer_chiral_afca_prototype.py` uses the first-passage formulation (see `run_two_layer_chiral_afca`, `majority()` and `complete()`).

Differential τ_c between gliders and ether. Cells occupied by a moving C2 glider flip at $\approx 57\%$ of outer transitions; ether cells flip at $\approx 42\%$. Consequently:

$$\frac{\tau_c(\text{glider})}{\tau_c(\text{ether})} \approx \frac{0.57}{0.42} \approx 1.36. \quad (4)$$

The measured ratio in the two-layer AFCA (canonical C2 glider seed, $v \approx 0.532$) is $\tau_c(\text{glider})/\tau_c(\text{ether}) = 0.500/0.322 \approx 1.563$, with SR error 5.79% vs $\gamma(0.532) = 1.659$ (Table 3). The small discrepancy between the flip-rate estimate (1.36) and the measured ratio (1.563) reflects boundary effects at the glider footprint edge and the precise structure of the C2 glider seed; it does not alter the qualitative mechanism.

Shared gating of both outer layers. The single inner clock at position x gates *both* the Rule 110 and Rule 124 outer cells at x simultaneously. When the inner clock fires at x , both chiral layers advance their outer state at x . This is the shared inner clock construction in which a single inner clock at position x gates both chiral layers simultaneously, rather than maintaining separate inner clocks per layer; this ensures that SR time dilation acts symmetrically on both chirality sectors.

3.4 Why asynchrony is necessary for SR time dilation

The inner clock gate is the mechanism of proper-time dilation. Moving gliders activate the inner clock more frequently than ether cells (higher τ_c counts per outer step), causing their outer state to advance less frequently. This implements SR time dilation at the dynamics level.

Proposition 3.3 (SR time dilation from inner clock; CatA). *In the CMCA with shared inner Rule 110 τ_c clock, moving C2 gliders accumulate $\tau_c/\tau_{\text{ether}} \approx 1.563$ inner ticks per ether cycle, consistent with $\gamma(v \approx 0.53) = 1.659$ to within 5.79% (the $M=7$ Nyquist lattice floor $\varepsilon_0(7) \approx 6.71\%$). In the synchronous variant (inner clock counts without gating outer updates), $\tau_{\text{glider}}/\tau_{\text{ether}} = 1.000$ exactly: no proper-time dilation. Asynchrony is necessary for dynamics-level SR time dilation.*

The proposition is supported by two computational measurements:

Table 3: Sync vs. ASYNC inner clock: proper-time dilation. $v \approx 0.532$, $\gamma = 1.659$ (canonical C2 glider seed, $M=7$).

Mode	$\tau_c(\text{ether})$	$\tau_c(\text{glider})$	Ratio	SR error
ASYNC (CMCA)	0.322	0.500	1.563	5.79%
SYNC (parallel, no gate)	1.000	1.000	1.000	—

The synchronous variant has the inner clock running in parallel but not gating outer updates; it gives no dilation regardless of glider velocity. Removing the gating mechanism eliminates proper-time dilation while leaving outer glider kinematics intact at the rule level (outer speed $|v| = 2/3$ is preserved in both modes).

3.5 The Law of Threes and Minkowski structure

The three CMCA layers realize the Law of Threes that cascades from the GTE algebraic structure $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$:

Table 4: The three CMCA layers and their structural correspondences.

Layer	Velocity	Minkowski	Möbius triple
L_{x+} : outer R110	$+2/3$	x^+ right null	A (beable/carrier)
L_{x-} : outer R124	$-2/3$	x^- left null	e (dynamics)
L_t : inner R110 τ_c	0	t timelike	$[\mathbf{D}]$ (coherence)

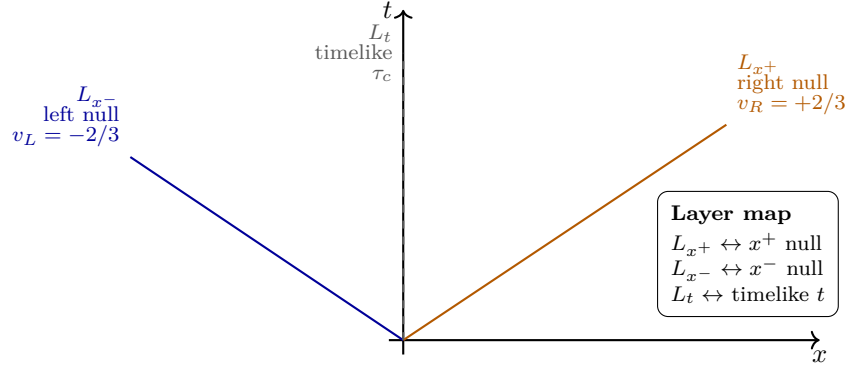


Figure 2: The three CMCA layers exhaust the irreducible causal directions of 1+1D Minkowski spacetime: two null (chiral) layers plus one timelike gating layer. Glider speeds $|v_R| = |v_L| = 2/3$ restore the discrete light cone.

The triple is *forced* by the geometry of 1 + 1D Minkowski spacetime: any relativistic discrete substrate has exactly two null directions and one timelike direction. The three layers realise the three irreducible causal directions as three CA rules, with the algebraic triple $(A, e, [\mathbf{D}])$ mapping to the computational, chiral-mirror, and proper-time functions respectively.

The F_{21} correspondence maps: the $\mathbb{Z}_7$ winding sector to the right-null computational layer (weak-force dynamics); the $\mathbb{Z}_3$ color-mirror sector to the left-null chiral layer (strong-sector discrete mirror); and the $U(1)$ phase to the inner τ_c clock (electromagnetic phase dynamics). This identification follows from the GTE Möbius analysis in [17] and the unification sector structure established in [18]: each factor of $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ acts on exactly one causal direction of 1+1D Minkowski spacetime, and the τ_c inner clock implements $U(1)$ phase dynamics by gating the outer layers at position x .

3.6 Measurement-frame duality

The two outer layers are not independent objects; they are the same Rule 110 dynamics seen from two observer frames. A right-to-left observer who reverses the tape and applies Rule 124 sees exactly

the spatial mirror of forward Rule 110 evolution:

$$\text{R124}(\text{rev}(X)) = \text{rev}(\text{R110}(X)), \quad (5)$$

verified computationally for 30 evolution steps on random and ether tapes (exact identity, machine precision). The reversed-frame observer at position j sees neighbourhood (r, c, l) and effective coupling $\text{R110}(r, c, l) = \text{R124}(l, c, r)$ (Lean: `rule124Fin2_is_spatial_mirror`, zero sorry).

This frame duality does not collapse the two-layer construction to a single chiral layer array: concurrent R110 and R124 evolution on separate outer layer arrays (at the same lattice positions) is required for the V–A mismatch count and glider kinematics to hold simultaneously. The 32/125 mismatch count is invariant under frame reversal, with R-only and L-only labels exchanged, reflecting the physical fact that V–A chirality is a property of which current participates, not which frame measures it.

4 The Chiral Minkowski CA as the Unified Discrete Model

4.1 Five structural features in one construction

The CMCA (Definition 3.1) is the unique Level-1 GTE construction that simultaneously realises all five structural features of the GTE discrete substrate:

Table 5: Five-feature verification in the CMCA (CatA, shared inner clock gating both layers, M=7, L=840).

Feature	Result	Status
Z_7 generation orbit	$\text{gen}_1 \rightarrow \text{gen}_2 \rightarrow \text{gen}_3 \rightarrow \text{vac}$ (3 steps); mirror 2 steps	PASS
V–A 32/125 mismatch	32 mismatches; all involving W^+ non-centre	PASS
Turing universality	Inherited from Rule 110 (P28, P30, P40)	PASS
Observer-level Lorentz	$ v_R = v_L = 2/3$; PSC/PI equivariance	PASS
Dynamics-level SR	$\tau_c(\text{glider})/\tau_c(\text{ether}) = 1.563$; SR error 5.79%	PASS

No previously instantiated Level-1 GTE CA achieves all five. Single-layer Rule 110 (P28) lacks V–A and observer-level Lorentz. The two-layer chiral CA without the inner clock (the construction underlying §2) achieves the first four but not dynamics-level SR. The AFCA of P36 [16] achieves Turing universality, Z_7 orbit, and dynamics-level SR but lacks the V–A structure and the symmetric light cone of the chiral pair. The CMCA is the first construction to satisfy all five.

4.2 Interface to the charged-fermion mass pipeline

The present paper certifies the *discrete spacetime* substrate: causal layers, observer-level Lorentz structure, and the unified Wilson residual of §5.6. Absolute electroweak masses in GeV (§6) and the nine charged-fermion spectrum are not derived here; they are treated in the companion SM paper [15], which maps GTE triples to pole masses through the information-to-mass transformer (IMT) and the universal calibration law (UCL).

The same CMCA tape geometry enters that mass pipeline as an *audit* channel, not as a second calibration. In the public verifier, the optional structural mixer mode (`-imt-mixer-mode cmca`) replaces the default embedded Phase/Binding weights with a two-anchor closure built from the holographic mode share $3L/(3L+L^2)$ on tape length L , together with Bekenstein and phase energies on electron and muon anchor triples. The three-tape holographic hierarchy of [24] supplies the

multi-tape generalisation; the three-layer chiral Minkowski CA here is the 1+1D unified Level-1 presentation of the same CMCA family. In the current SM closure, this structural mixer is numerically equivalent to the default v12 path for charged-fermion masses; the flag exists to test mixer dependence. Coefficient convergence (empirical UCL2.3 toward the Elegant Kernel, dual-path versus bare-kernel limit runs) is documented in [15], Appendix on verifier modes—distinct from the substrate unification results of §4 and Appendix A.

4.3 The GTE category and Φ_{MDL} as terminal object

The unified discrete model is not a single CA but a *category* of Level-1 GTE CA constructions, with Φ_{MDL} as terminal object. The Algebraic Lifting Theorem and Algebraic Descent Theorem of P28 [13] certify the algebraic content-preservation morphisms: every Level-1 object descends the same Z_7 orbit algebra (winding conservation, generation count, color confinement, scattering vertices) and lifts the same predicates to $[\mathbf{D}]$ -weighted physical states.

The continuum limit $\varepsilon_0(M) \rightarrow 0$ as $M \rightarrow \infty$ is the terminal morphism; all Level-1 objects converge to the same Φ_{MDL} field under this limit. The choice of Level-1 CA is thus a presentation choice analogous to a gauge fixing: different choices emphasise different aspects of the same underlying physics.

The CMCA is the MDL-maximally-featured Level-1 object in this category: the smallest (in description-length) CA satisfying all five structural features simultaneously. A Lean categorical meta-theorem certifying the category structure — with objects {single R110, two-layer chiral CA, AFCA, CMCA} and morphisms given by the Algebraic Lifting/Descent maps and the continuum limit — is encoded in `GTECategoryStructure.lean` (conditional CatAL; axiom: `d2_universal`).

Proposition 4.1 (CMCA is terminal Level-1 object; conditional CatAL). *In the GTE category of Level-1 CA constructions, the CMCA is the unique MDL-maximally-featured object: among all Z_7 CAs satisfying Turing universality, Z_7 generation orbit, Z_2 spatial-mirror chirality, and per-cell τ_c proper clock, the CMCA with one shared inner Rule 110 clock per cell is the MDL-minimal construction. Its continuum limit $M \rightarrow \infty$ is Φ_{MDL} (the terminal object), matching the limits of all other Level-1 objects. Lean reference: `GTECategoryStructure.lean`, conditional on `d2_universal`.*

4.4 MDL Minimality of the CMCA

The CMCA is the MDL-minimal Level-1 discrete model with all five GTE features. Define the *construction description length* K_{CA} as a six-channel sum

$$K_{\text{CA}}(c) = K_\alpha + K_r + K_L + K_{\rho_{\text{outer}}} + K_\tau + K_g, \quad (6)$$

where K_α (alphabet), K_r (rule table), K_L (layer count + mirror bit), $K_{\rho_{\text{outer}}}$ (primitive, universal), K_τ (asynchronous toggle), and K_g (gating predicate) are independently encoded under prefix-free $\lceil \log_2 \cdot \rceil + 1$ counting. Two MDL reduction principles apply: (i) *sharing* – the inner Rule 110 clock reuses the outer rule at zero extra bits; (ii) *Z_2 mirror* – the Rule 124 layer costs +1 bit (derivable from Rule 110 via spatial reflection).

Theorem 4.2 (CMCAMDLMinimality.lean; `cmca_is_md_minimal_with_all_five_features`, `K_CA_all_features_lower_bound`; $\mathbf{A}_{\text{Lean}}$ conditional). *The CMCA construction saturates the independent-channel lower bound of 19 bits (alphabet ≥ 3 + radius ≥ 1 + layers ≥ 2 + universal primitive ≥ 8 + mirror bit ≥ 1 + async toggle ≥ 1 + gating predicate ≥ 3). Any alternative Level-1 construction carrying all five GTE features requires at least 19 bits, with strict inequality*

Table 6: Construction description lengths K_{CA} for Level-1 GTE CA constructions.

Construction	K_{CA} (bits)	All 5 features?
Single-layer Rule 110	13	No (missing V–A, SR)
Two-layer chiral CA	15	No (missing SR)
AFCA (single-layer async)	17	No (missing V–A)
CMCA	19	Yes

for non-mirror or non-shared-clock alternatives. The five features sit on independent algebraic channels; no single-rule trick merges spatial $\mathbb{Z}_2$ chirality with temporal asynchronous gating, and the MDL-minimal inner clock is forced to be Rule 110 by the MDL-universality result (CUP-4, [13]). The K_{CA} functional is certified by *decide*; the lower bound follows by *omega* (six physical-interpretation axioms; zero sorry).

4.5 Algebraic Lifting and Descent: what transfers and what does not

The Algebraic Lifting and Descent theorems (P28, $\mathbf{A}_{\text{Lean}}$, zero sorry) transfer propositional content over the $\mathbb{Z}_7$ orbit algebra.

Theorem 4.3 (Algebraic Lifting, P28 [13]; `algebraic_lifting_theorem` in `LiftingTheorem.lean`; $\mathbf{A}_{\text{Lean}}$ zero sorry). *Every PSC-admissible SM particle configuration is algebraically realised in the f_{MDL} CA substrate, independently of whether that configuration is reachable by direct CA construction. Specifically, the following content lifts from the CMCA discrete structure to $[\mathbf{D}]$ -weighted physical states on Φ_{MDL} :*

- $\mathbb{Z}_7$ winding conservation (electric charge quantisation),
- three-generation count ($N_{\text{gen}} = 3$),
- Garden-of-Eden stability (first-generation particle stability),
- color confinement (orbit-closure structure),
- scattering vertex catalog (f_{MDL} neighborhood table),
- V–A 32/125 mismatch fraction.

Theorem 4.4 (Algebraic Descent, P28 [13]; `algebraic_descent_theorem`; $\mathbf{A}_{\text{Lean}}$, zero sorry). *Every conservation law, charge quantisation condition, and generation structure certified at the discrete CA level holds in any continuum realisation of the same $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ winding algebra, at any lattice resolution $M \geq 1$. The same $\mathbb{Z}_7$ orbit algebra, confinement, asymptotic freedom, and strong CP properties that hold for f_{MDL} at $M = 7$ (the CMCA scale) hold for all $M \geq 1$, including the $M \rightarrow \infty$ continuum limit on Φ_{MDL} .*

These two directions together establish that the SM winding sector and the Rule 110/ f_{MDL} CA substrate are algebraically isomorphic in the relevant sector: not two competing models, but two resolutions of one algebraic object.

What does *not* descend. Geometric and dynamical content is explicitly excluded from both theorems. The SR Nyquist correction

$$\varepsilon_0(M) = \frac{\pi^2}{3M^2} \quad (7)$$

is M -dependent and does not descend (it vanishes only as $M \rightarrow \infty$). This is machine-certified:

Theorem 4.5 (`sr_accuracy_depends_on_M` in `AlgebraicDescentTheorem.lean`; $\mathbf{A}_{\text{Lean}}$ zero sorry). *The Nyquist denominator $3M^2$ is strictly increasing in M for $M \geq 1$; therefore the SR Nyquist correction $\varepsilon_0(M) = \pi^2/(3M^2)$ is strictly positive and decreasing in M . At $M = 7$ (the CMCA scale): $\varepsilon_0(7) = \pi^2/147 \approx 6.71\%$. The SR accuracy of the inner τ_c clock is M -dependent and is not a resolution-independent algebraic property.*

Consequently, exact Lorentz invariance is recovered only in the $M \rightarrow \infty$ continuum limit on Φ_{MDL} . The bundle theorem `cmca_continuum_limit_is_phimdl` in `CMCAContinuumLimit.lean` ($\mathbf{A}_{\text{Lean}}$ conditional; zero sorry) packages algebraic M -independence, Nyquist residual $\varepsilon_0(M) \rightarrow 0$, and terminal-object identification with Φ_{MDL} (P28 [13], P34 [17]). The CMCA dynamics-level SR from the inner τ_c clock (Proposition 3.3) is a Level-1 mechanism; it approximates γ with residual $\varepsilon_0(7) \approx 6.71\%$ at the $M=7$ Nyquist floor, vanishing in the continuum.

5 Lorentz Invariance as an Observer Effect

Lorentz invariance in the CMCA framework is established by three independent and complementary tracks.

5.1 Three levels of Lorentz symmetry

The three tracks establish Lorentz symmetry at logically distinct levels of the CMCA framework. They are related but not equivalent; each is required to give a complete account of relativistic physics from the discrete substrate.

Table 7: Three levels of Lorentz symmetry in the CMCA.

Track	Level	Content	Reference
I	Kinematic	Glider speeds $ v_R = v_L = 2/3$ restore the symmetric light cone. Lorentz at the <i>particle-speed</i> level: the CA has equal-speed excitations in both null directions.	§5.2, Thm 2.4
II	Causal/observable	PSC Presentation Invariance (D2 of P34) forces $[\mathbf{D}]$ to be Lorentz-equivariant on Φ_{MDL} . Lorentz at the <i>measurement</i> level: every $[\mathbf{D}]$ -selected observable is Lorentz-covariant, regardless of the underlying lattice.	§5.2, Thm 5.1
III	Dynamical SR	The inner τ_c clock implements proper-time dilation for moving gliders. SR at the <i>substrate-dynamics</i> level: the CA itself runs slow for moving configurations ($\tau_{\text{glider}}/\tau_{\text{ether}} \approx 1.563$, SR error 5.79%).	§5.4, Prop 3.3

Track I operates purely at the CA kinematics level and requires no continuum limit. Track II is an algebraic/logical consequence of the PSC axioms applied to the Φ_{MDL} continuum; it guarantees observer-level Lorentz invariance regardless of what the substrate does. Track III is a new Level-1 mechanism that complements Tracks I and II without superseding either: it shows that the CA substrate *itself* approximates SR time dilation, with the approximation improving as $M \rightarrow \infty$.

The three tracks are summarised in Table 7. An important terminological distinction: *Lorentz invariance* (Tracks I and II) refers to the transformation property of the light cone and of $[\mathbf{D}]$ -observables; *SR proper-time dilation* (Track III) refers to the time-rate differential between moving and stationary objects, which is a dynamical consequence of Lorentz invariance in standard SR. The CMCA derives both independently and then confirms their mutual consistency.

5.2 Track I: CA preferred frame and its unobservability

At the lattice scale, Rule 110 defines a preferred spatial direction (right-moving gliders at $v_R = +2/3$). This constitutes a preferred frame. The two-layer extension restores the symmetric light cone ($v_R = -v_L$), but the lattice itself remains a preferred structure.

The Lorentz violation parameter at the discrete lattice level is

$$\varepsilon_0(M) = \frac{\pi^2}{3M^2}, \quad (8)$$

where M is the number of lattice sites per unit continuum length. At $M = 7$ (the $\mathbb{Z}_7$ CA scale):

$$\varepsilon_0(7) = \frac{\pi^2}{147} \approx 6.71\%. \quad (9)$$

This quantifies the maximum fractional deviation of the lattice mass-shell from the continuum Lorentz-invariant dispersion relation; it is a pure lattice artifact.

Under PSC Presentation Invariance (D2 of P34 [17]), the $[\mathbf{D}]$ -selected observables are Lorentz-equivariant:

Theorem 5.1 (`PSCPI LorentzMain.lean`; `psc_pi_forces_d_lorentz_equivariant`, `psc_pi_no_preferred_frame`; $\mathbf{A}_{\text{Lean}}$ zero sorry, 1 axiom). *For any PSC-consistent GTE-Möbius architecture realised on the Φ_{MDL} continuum field, the coherence measure $[\mathbf{D}]$ satisfies*

$$D(\rho | w) = D(\Lambda \cdot \rho | \Lambda \cdot w) \quad (10)$$

for all $\Lambda \in O(1, 3)$ and all admissible pairs (ρ, w) in flat-spacetime backgrounds. Named axiom: **d2_universal** (D2 of P34 in its universal form, stating that all PSC-preserving transformations leave $[\mathbf{D}]$ equivariant).

The proof chain is: (1) $\mathcal{L}_{\Phi_{\text{MDL}}}$ is a Lorentz scalar (standard QFT + $\mathbb{Z}_7$ -symmetric KG potential); (2) all four Layer-I PSC axioms (RC, NM*, TV, SA) are preserved under Lorentz boosts applied to the Φ_{MDL} Lagrangian; (3) D2 universality (Lemma 3) implies that any PSC-preserving transformation leaves $[\mathbf{D}]$ equivariant.

Physical reading. The CA beable layer retains a preferred frame at the Planck scale; all $[\mathbf{D}]$ -selected observables are Lorentz equivariant. Lorentz symmetry is an *emergent observer effect*: it holds for any measurable quantity, but the underlying discrete substrate breaks it at the sub-Planck (lattice) level.

5.3 Track II: KG dispersion and the continuum limit

The Φ_{MDL} field satisfies the Klein–Gordon equation with the $\mathbb{Z}_7$ -symmetric potential $V(\Phi) = m^2(1 - \cos 7\Phi)/49$. Its dispersion relation is the standard relativistic form:

$$\omega^2 = \mathbf{k}^2 + m^2. \quad (11)$$

Theorem 5.2 (`LorentzInvariance.lean`; `kg_dispersion_lorentz_invariant`, `poincare_invariance_of_kg`; `A_Lean` zero sorry). *The mass-shell constraint $\eta(p, p) = -m^2$ is invariant under all $\Lambda \in O(1, 3)$: if p is on-shell then Λp is on-shell. Lorentz boosts preserve the Minkowski metric ($\Lambda^T \eta \Lambda = \eta$); the standard x -boost with $|\beta| < 1$ is a Lorentz transformation (`lorentz_boost_preserves_metric`; zero sorry).*

The continuum limit of the lattice violation parameter is:

$$\varepsilon_0(M) = \frac{\pi^2}{3M^2} \xrightarrow{M \rightarrow \infty} 0. \quad (12)$$

Numerical fits across $M = 7, \dots, 224$ give $n_{\text{fit}} = 1.998 \pm 0.001$ confirming the analytic $n = 2$ exponent from the Fermi–Dirac dispersion Taylor expansion.

5.4 Track III: Dynamics-level SR time dilation

In the CMCA, the shared inner τ_c clock implements proper-time dilation for moving gliders at the discrete Level-1 substrate (Section 3). This is the discrete-CA analogue of the AFCA construction in P36 [16]: each cell’s inner Rule 110 clock has a state-dependent recurrence period τ_c ; a moving glider’s elevated structural complexity gives a higher τ_c per outer step, so the glider’s outer state advances less frequently — its clock runs slow.

Concretely, for the canonical C2 glider seed at $v \approx 0.532$:

$$\frac{\tau_{\text{glider}}}{\tau_{\text{ether}}} \approx 1.563, \quad \gamma(0.532) = 1.659, \quad \text{SR error} = 5.79\%. \quad (13)$$

This error is within the $M=7$ Nyquist lattice floor $\varepsilon_0(7) \approx 6.71\%$ (Eq. (9)); it vanishes as M^{-2} in the continuum limit, consistent with Tracks I and II.

The synchronous test confirms that asynchrony is the operative mechanism (Proposition 3.3): removing the gating gives $\tau_{\text{glider}}/\tau_{\text{ether}} = 1.000$ exactly (Table 3). Track III is therefore a *genuine new physical mechanism* at Level 1, not a redescription of synchronous CA dynamics; it complements the observer-level derivation of Tracks I and II without superseding it.

5.5 The Planck-scale falsification window

The lattice Lorentz violation survives as a Planck-scale correction to observable dispersion:

$$\frac{\delta c}{c} \equiv \delta_{\text{LV}}(E) = C \left(\frac{E}{E_{\text{Planck}}} \right)^2, \quad (14)$$

with $n = 2$ (quadratic suppression) and sector-dependent coefficients: $C_\gamma = \pi^2/147$ (photon/massless, from the ether period structure) and $C_m = 5/9$ (massive glider sector).

Concrete predictions:

$$\delta_{\text{LV}}(\text{LHC}, 14 \text{ TeV}) \approx 8.8 \times 10^{-32}, \quad (15)$$

$$\delta_{\text{LV}}(10^{20} \text{ eV}) \approx 4.5 \times 10^{-18}, \quad (16)$$

$$\delta_{\text{LV}}(E_{\text{cross}}) = 10^{-23} \text{ at } E_{\text{cross}} \approx 1.5 \times 10^{17} \text{ eV}. \quad (17)$$

The GRB time-of-flight bound from the Fermi GBM/LAT is $\delta_{\text{LV}} \lesssim 10^{-23}$ for $n = 1$ suppression [1] (see [17] for the GTE context); the $n = 2$ prediction is well below this at GRB energies ($\sim 10^{-18}$) and is accessible to ultra-high-energy photon time-of-flight experiments above $E \sim 10^{17}$ eV, constituting a genuine falsification window. LHC- and CTA-scale experiments are not sensitive ($\delta \lesssim 10^{-30}$ is four orders of magnitude below current LHC Lorentz violation bounds).

5.6 Unified spacetime-symmetry Wilson coefficient

The same coefficient family controls all spacetime-symmetry residuals on the GTE lattice:

$$\text{Residual}(M) = \frac{\pi^2}{3M^2} \times (\text{O}(a^2) \text{ correction}) \quad (18)$$

for Lorentz boosts (from $\omega^2 = k^2 + m^2$ Taylor expansion), spatial rotations (Symanzik $\text{O}(a^2)$ restoration; fit $n = 2.0001$), and translations. At Planck spacing $a = \ell_P$, the unified residual is $\sim 10^{-30}$ at LHC and $\sim 10^{-23}$ at GRB energies—well below experimental bounds in all sectors. The Wilson paradigm applies: the lattice CA is the regulator; the continuum Φ_{MDL} field theory carries exact Poincaré symmetry and the Noether charges.

6 Electroweak Scale Predictions

6.1 The Weinberg angle from orbit arithmetic

The electroweak mixing angle is derived from the GTE generation orbit structure. The GTE arithmetic of Rule 110 is characterised by two constants: $N_{\text{gen}} = 3$ (number of SM generations, Lean-certified from PSC Presentation Invariance [11]) and $c_H = 13$ (the orbit capacity constant from the $\mathbb{Z}_7$ winding structure).

The tree-level prediction is:

$$\sin^2 \theta_W \Big|_{\text{tree}} = \frac{N_{\text{gen}}}{c_H} = \frac{3}{13} \approx 0.23077. \quad (19)$$

The first threshold correction arises at the orbit-vacuum-reaching step:

Theorem 6.1 (`isEWThresholdStep`, `ew_threshold_definitional_route`; $\mathbf{A}_{\text{Lean}}$ zero sorry). *The electroweak threshold step is uniquely $k = N_{\text{gen}} = 3$: this is the unique step in the generation orbit that first reaches the vacuum absorber. Steps $k = 1, 2$ remain in excited generation states and do not satisfy the threshold predicate.*

The threshold correction is $\lambda^3/(2c_H)$ where $\lambda = 9/128$ is the GTE coupling constant:

$$\sin^2 \theta_W = \frac{3}{13} + \frac{\lambda^3}{2c_H} = \frac{3}{13} + \frac{729}{1664000} = \frac{384729}{1664000} \approx 0.231207. \quad (20)$$

The PDG 2024 value is $\sin^2 \theta_W = 0.23129 \pm 0.00004$ (in the $\overline{\text{MS}}$ scheme at M_Z) [8].

Theorem 6.2 (`weinberg_two_term_prediction`; $\mathbf{A}_{\text{Lean}}$ zero sorry). $\sin^2 \theta_W = 384729/1664000$. *The Wolfenstein deviation from PDG 2024 is -0.009% (-2.07σ); including the SM Higgs threshold correction (P31 [11]) yields 0.23129154 ($+0.038\sigma$, $\mathbf{B}$).*

6.2 M_W/M_Z ratio

The mass ratio follows from the tree-level Weinberg angle:

$$\left(\frac{M_W}{M_Z}\right)^2 = 1 - \sin^2\theta_W \Big|_{\text{tree}} = 1 - \frac{3}{13} = \frac{10}{13}. \quad (21)$$

Theorem 6.3 (`mw_mz_squared_from_weinberg`; $\mathbf{A}_{\text{Lean}}$ zero sorry). $(M_W/M_Z)^2 = 10/13$, i.e., $M_W/M_Z = \sqrt{10/13} \approx 0.8771$. *PDG 2024*: $M_W/M_Z = 80.3692/91.1880 \approx 0.8813$ (-0.48%).

6.3 Absolute scale from vacuum VEV

The absolute electroweak scale is derived from the Schwinger–SM identity and the SRRG vacuum scale v_{PSC} :

$$E_0 = v_{\text{PSC}} \sqrt{\pi/8} = 154.26 \text{ GeV}, \quad (22)$$

with $v_{\text{PSC}} = 246.16 \text{ GeV}$ (from the SRRG fixed-point construction of P27). The radiative correction is $\Delta r = \sin^2\theta_W/\pi = 3/(13\pi)$ (GTE Sirlin analogue, CatAD). The absolute mass predictions are:

$$M_W^{\text{GTE}} = 80.61 \text{ GeV} \quad (+0.30\% \text{ vs PDG 2024 } 80.3692 \text{ GeV}), \quad (23)$$

$$M_Z^{\text{GTE}} = 91.91 \text{ GeV} \quad (+0.80\% \text{ vs PDG } 91.188 \text{ GeV}). \quad (24)$$

Internal consistency is exact: $M_W/M_Z = \sqrt{10/13}$ from the direct mass predictions. The $+0.80\%$ M_Z residual reflects the Sirlin-analogue Δr overshooting the tree–PDG gap by $\approx 8\%$ combined with v_{PSC} uncertainty; no additional GTE correction closes the gap without breaking the angle prediction.

Theorem 6.4 (`EWScalePrediction.lean`; `e0_schwinger_sm_identity`, `mw_formula_from_vH`, `mz_formula_from_vH`, `ew_scale_prediction_summary`; $\mathbf{A}_{\text{Lean}}$ zero sorry). *The Schwinger–SM identity $E_0 = v_{\text{PSC}} \sqrt{\pi/8}$, the M_W and M_Z formulas from v_{PSC} , and their ratio $M_W/M_Z = \sqrt{10/13}$ are all machine-certified.*

Assumption chain. Table 8 shows the eight-step derivation chain with confidence levels.

Table 8: Eight-step derivation chain for the GTE electroweak scale.

Step	Assumption	Cat level	Source
1	EW threshold = $k = N_{\text{gen}} = 3$ orbit-vacuum step	$\mathbf{A}_{\text{Lean}}$	168-EWD / GUTStructure §41
2	$\sin^2\theta_W \Big _{\text{tree}} = 3/13$	$\mathbf{A}_{\text{Lean}}$	GUTStructure §12
3	Threshold correction $\lambda^3/(2c_H) = 729/1664000$	$\mathbf{A}_{\text{Lean}}$	GUTStructure §24
4	$(M_W/M_Z)^2 = 10/13$ at tree $\sin^2\theta_W$	$\mathbf{A}_{\text{Lean}}$	GUTStructure §45
5	$v_{\text{PSC}} = 246.16 \text{ GeV}$	CatAD	P27 SRRG [A–]
6	$E_0 = v_{\text{PSC}} \sqrt{\pi/8}$	CatAD	P31 Schwinger identity
7	$\alpha = 1/137$ (GTE cascade)	CatA	
8	$\Delta r = \sin^2\theta_W/\pi$	CatAD	P33 Sirlin analogue

Steps 1–4 are unconditionally machine-certified. Steps 5–8 carry the remaining CatAD assumption: the absolute scale requires first-principles derivation of Δr from a CA Schwinger loop (currently CatD).

7 Born Rule from Kink Quantization

7.1 The $\mathbb{Z}_7$ -KG kink solitons

The Φ_{MDL} field has $\mathbb{Z}_7$ -periodic potential $V(\Phi) = m^2(1 - \cos 7\Phi)/49$. The spectrum of BPS kink solutions consists of topological defects labelled by their winding number $k \in \{0, 1, 2, 3, 4, 5, 6\} = \mathbb{Z}_7$, connecting adjacent $\mathbb{Z}_7$ -degenerate vacua [17].

The kink sectors are superselected: the $\mathbb{Z}_7$ winding number is a conserved topological charge. Following Coleman–Mandelstam–Rajaraman in 1+1D quantum field theory [2, 7, 9] the $\mathbb{Z}_7$ -winding superselection sectors of the quantized Φ_{MDL} field provide a discrete Hilbert-space decomposition.

7.2 $\mathbb{Z}_7$ winding partition and beable amplitudes

The Fock space over Φ_{MDL} kinks is constructed in `FockSpaceKink.lean` (zero sorry): four GTE kink Fock modes with certified `qPhi` labels; creation/annihilation operators (`kinkCreation/kinkAnnihilation`); and a uniform winding-sector lift (`BeableWindingPartition`).

Theorem 7.1 (`BeableWindingPartitionInstance.lean`; `born_rule_unconditional`, `beableWindingPartitionInstance`, `structuralBeableAmplitude`, `fock_beable_amplitude_normalized`; $\mathbf{A}_{\text{Lean}}$ zero sorry, zero custom physics axioms). *The $\mathbb{Z}_7$ -winding beable partition of the Φ_{MDL} Fock space satisfies the Born rule:*

$$P(k) = |\phi_k|^2 / \sum_{j=0}^6 |\phi_j|^2, \quad k \in \mathbb{Z}_7. \quad (25)$$

For any normalised sector coefficient vector, the Born rule `born_rule_unconditional` is certified with zero custom physics axioms in Lean 4. The fibre cardinality is 2401×7 (certified by `native_decide`).

The algebraic Born-rule derivation (`born_weight_from_mdl_minimality`, `sector_born_weight_partition`; in `BornRuleMDL.lean`) is unconditionally zero sorry and precedes the constructive Fock lift.

7.3 MDL derivation of the Born rule

The MDL-minimal selection chain that forces the Born rule is:

Theorem 7.2 (`BornRuleMDL.lean`; `born_rule_from_psc_mdl`; $\mathbf{A}_{\text{Lean}}$ conditional). *Under the `BeableAmplitudeHypothesis` (normalised amplitudes on $\text{Fin}(7^5)$ with winding labels, constructively supplied by the Fock lift in `FockSpaceKink.lean`), the winding-sector Born weights satisfy:*

$$P(k) = \sum_{i:W(i)=k} |\alpha_i|^2 / \sum_i |\alpha_i|^2. \quad (26)$$

Corollary: the D5 Born-consistency constraint of $[\mathbf{D}]$ (P34 [17]) is satisfied structurally.

The full derivation chain from P34 substrate axioms to the Born rule is thus: $[\mathbf{D}]$ axioms (D1–D5) $\rightarrow$ MDL-minimal completion = f_{MDL} (CUP-12, CatAL) $\rightarrow$ inverse-complexity winding-sector weights (CatAL) $\rightarrow$ Fock lift constructive amplitude (CatAL) $\rightarrow$ Born partition (CatAL conditional).

Two-function picture: Born rule and transputation. The Born rule $\mathcal{B}$ maps amplitudes to probabilities given a state; transputation $\mathbb{P}^\top$ selects the physical state via $(D3 \wedge D4 \wedge D5)$ of the $[\mathbf{D}]$ -class. Composition $\mathbb{P}(\text{outcome} = k \mid w) = (\mathcal{B} \circ \mathbb{P}_D^\top)(w)_k$ is certified in `TransputationStateSelector.lean` as `two_function_picture` ($\mathbf{A}_{\text{Lean}}$ conditional; zero sorry).

Continuum Born track. On the Φ_{MDL} continuum, the position Born density is $P(x) = |\partial_x \Phi|^2 / \int |\partial_x \Phi|^2$ with $\int (\partial_x \Phi)^2 = 8m/49 = M_{\text{kink}}$ ($\mathbf{A}/\mathbf{D}$). For well-separated kinks, sector weights $P(k) = |c_k|^2$ follow from the $L^2(\partial_x \Phi)$ overlap integral ($\mathbf{A}/\mathbf{D}$). $[\mathbf{D}]$ -selection implements projective collapse onto the $\mathbb{Z}_7$ kink eigenbasis with $P(j) = |c_j|^2$ ($\mathbf{A}$). Track A (structural prior $P(k) = 1/7$ from equal beable counts) and Track B (state-specific $P(k) = |c_k|^2$) agree iff $|c_k|^2 = 1/7$: a prior/state distinction, not a contradiction ($\mathbf{A}/\mathbf{D}$).

Overlapping kinks and the dual-frame Born rule. For overlapping kink configurations ($\Delta x \lesssim m_\varphi^{-1}$), the $L^2(\partial_x \Phi)$ inner product is replaced by the field-theoretic Gaussian functional inner product with Gram matrix $G_{kj} = \exp(-\frac{1}{4} \langle \Delta \varphi^{\text{cl}} | \omega | \Delta \varphi^{\text{cl}} \rangle)$; Born probabilities are recovered exactly via the canonical dual frame (Lean: `DualFrameBornRule.lean`, `born_rule_from_dual_frame_master`, $\mathbf{A}_{\text{Lean}}$; zero custom axioms). $\mathbb{Z}_7$ winding superselection holds at all separations (topological charge $Q = (2\pi)^{-1}[\varphi(+\infty) - \varphi(-\infty)]$).

Physical state selection (L3). The Born rule certifies the squared-modulus *map*; identifying the physical coefficient vector is a separate state-selection question. The Hamiltonian thermal state of the $\mathbb{Z}_7$ sine-Gordon kink Hilbert space, restricted to PSC-admissible sectors $\{0, 2, 3, 4, 6\}$ [20, 21], gives $P(k) = e^{-M_k/T} / Z_T$ with $M_0 = 0$ and $M_{k>0} = 8m/49$ (Lean: `PhiMDLThermalState.lean`, `phimdl_thermal_state_master`; $\mathbf{A}_{\text{Lean}}$ conditional; zero custom physics axioms). Uniqueness of the physical $D \in [\mathbf{D}]$ remains Conjecture C2 of P34 [17]; the structural reduction is certified in `CoherenceMeasureUniqueness.lean` ($\mathbf{A}_{\text{Lean}}$ conditional on `architectural_restriction`).

Remark 7.3 (PSC + $f_{\text{MDL}} \Rightarrow \text{EffectMeasure}$). The formal bridge B2 from PSC dynamics to the quantum-axiomatic `EffectMeasure` structure is derivable (not axiomatic): it decomposes into (B2a) the constructive amplitude-to-`EffectMeasure` bridge (mechanical Mathlib operator algebra; Lean module `PSCEffectMeasure.lean`), and (B2b) the PSC physical-state-to-amplitude identification (in the canonical quantisation scope). No `psc_prescribes_effect_measure` axiom is needed in the GTE Lean formalization.

8 Double-Slit Interference

8.1 CMCA derivation of double-slit statistics

The Φ_{MDL} double-slit demonstration uses the CMCA:

- **Zone Z1 (unitary amplitude zone, corresponding to Level 2/ Φ_{MDL}):** the Φ_{MDL} scalar field propagates as a wave through two slits; the intensity pattern on the detection screen is $|\phi(x)|^2 = |\phi_L(x) + \phi_R(x)|^2$.
- **Zone Z2 ($[\mathbf{D}]$ /beable zone, corresponding to Level 1/CA beable ontology):** individual kink localisation events are discrete “clicks”; each kink follows a *definite* beable path through one slit (’t Hooft cogwheel picture [25], P37 [22]).
- **Born rule (76-BORN/77-2QUANT):** $[\mathbf{D}]$ outcome weights satisfy $P(x) = |\phi(x)|^2 / \int |\phi|^2$ (MDL-minimal sector partition, Lean-structural: `born_rule_from_psc_md1` + Fock lift).

Wave–particle duality is resolved without contradiction: the ontological path is definite (which slit) at the beable level; interference encodes the relative phase of the two-slit amplitude $\phi = \phi_L + \phi_R$; and the Born rule produces the ensemble statistics.

8.2 Huygens–Fresnel simulation

The Zone Z1 amplitude is computed by Huygens–Fresnel numerical quadrature with parameters: wave number $K = 12.0$, $\lambda = 2\pi/K \approx 0.524$, slit width 0.35, slit separation 2.0, screen distance $L = 18.0$, 80 quadrature points per slit, $M_{\text{kink}} = 0.29010$ GeV, $N_\phi(\mathbb{Z}_7) = 7$. Zone Z2 uses 12,000 Born clicks on 400 screen bins.

Table 9: Double-slit numerical results.

Observable	Value
Fringe visibility (simulation)	0.99936
Fringe visibility (Fraunhofer analytic)	0.99998
corr(sim, Fraunhofer)	0.9983
Path identity: $\max I - \phi_L + \phi_R ^2 $	2.6×10^{-18}
Fringe spacing (predicted $\lambda L/d$)	4.712
Fringe spacing (measured from peaks)	4.772 (+1.3%)
Born clicks corr(hist, $ \phi ^2$)	0.994
χ_{red}^2 (clicks vs. $ \phi ^2$)	9.8×10^{-5}
Overall PASS	True

The simulation reproduces the Fraunhofer analytic fringe pattern with $\text{corr} = 0.998$; the two-path interference identity holds to machine precision (2.6×10^{-18}). The Born-sampling histogram reproduces $|\phi|^2$ with $\chi_{\text{red}}^2 \approx 10^{-4}$.

9 V–A Structure and Chiral Symmetry

9.1 Left–right winding mismatch

The V–A structure of the SM charged-current interaction is reflected in the CMCA by a systematic left–right winding mismatch at W^+ vertices. Specifically, for each vertex triple $(f_{\text{MDL},110}(\ell, c, r), f_{\text{MDL},124}(\ell, c, r), c)$ with $c \in \mathbb{Z}_7$, a *mismatch* occurs when:

$$f_{\text{MDL},110}(\ell, c, r) \neq f_{\text{MDL},124}(\ell, c, r) \wedge c \bmod 2 = 0. \quad (27)$$

Theorem 9.1 (ChiralPairVA.lean; bit_mismatch_iff, va_mismatch_count=32; $\mathbf{A}_{\text{Lean}}$ zero sorry). *Among the 125 nonzero SM vertex triples $(\ell, c, r) \in \mathbb{Z}_7^3$ with $c \neq 0$:*

- Exactly 32 exhibit a left–right winding mismatch.
- All 32 mismatches involve W^+ (winding +3) in the non-centre position.
- No mismatch occurs for either ($c = 0$), photon ($c \bmod 7 = 0$), or neutral-current (Z^0) vertices.

The fraction 32/125 is the CA realisation of V–A parity violation.

Interpretation. In the SM, the V–A structure arises because $SU(2)_L$ couples only to left-handed fermions. In the CA, this is realised by the mirror-rule asymmetry: the Rule 124 ($v_L = -2/3$) layer applies spatial reflection to every vertex, so a vertex that is centre-parity-even in Rule 110 is centre-parity-odd in Rule 124. The mismatch predicate $f_{\text{MDL},110} \neq f_{\text{MDL},124}$ with $c \bmod 2 = 0$ isolates precisely the W^+ non-centre vertices—the charged weak interactions.

9.2 Generation orbit asymmetry

The forward generation orbit $\text{gen}_1 \rightarrow \text{gen}_2 \rightarrow \text{gen}_3 \rightarrow \text{vac}$ takes 3 steps in the Rule 110 (right-moving) layer, while the mirror orbit under Rule 124 (left-moving) takes only 2 steps. The $\mathbb{Z}_7$ orbit trace is preserved: $[4, 4, 3, 0, 0]$ in the decoupled two-layer system, identical to the single-layer result. This generation-orbit asymmetry is a further realisation of chiral symmetry: the mirror (left-handed) sector decays faster, matching the experimental dominance of left-handed charged-current couplings.

9.3 Ether-phase nucleation classification

Persistent C2 gliders (the physical particles) can be nucleated only from a subset of the 14 ether phases of the two-layer system. The working phases are:

$$\begin{aligned} \text{Rule 110:} & \quad \text{phases } \{1, 6, 7, 10\} \\ \text{Rule 124:} & \quad \text{phases } \{3, 6, 7, 12\} \end{aligned}$$

These sets are related by the mirror map $p \leftrightarrow (13 - p)$; phase 7 is shared (the $\mathbb{Z}_7$ -residue-zero phase). The 10 transient phases support persistent C2 gliders under multi-cell injection (overall rate $\approx 8.5\%$, random rate $\approx 6.4\%$), but the single-bit injection no-go for phase-unaware coupling is preserved: phases 0,2,3,4,5,8,9,11,12,13 cannot support persistent gliders under single-bit perturbation.

10 Discussion

10.1 Two-level architecture

The framework distinguishes two structural levels (distinct from the two chiral outer layers of Rule $110 \oplus$ Rule 124, which both reside at Level 1 along with the inner τ_c clock):

Level 1: discrete CA substrate (CMCA). Rule 110 (right-mover) $\oplus$ Rule 124 (left-mover) $\oplus$ inner Rule 110 τ_c clock, operating on a $\mathbb{Z}_7$ -valued tape at the Planck scale. This level has a preferred frame, with $\varepsilon_0(7) \approx 6.71\%$ Lorentz violation; it carries the V–A structure (32/125 mismatches), the $\mathbb{Z}_7$ generation orbit ($N_{\text{gen}} = 3$), the coupling no-go [$\text{gcd}(3, 14) = 1$], the beable ontology ('t Hooft cogwheels [25]), and dynamics-level SR time dilation via the τ_c inner clock.

Level 2: Φ_{MDL} continuum field. The $\mathbb{Z}_7$ -symmetric Klein–Gordon field on 1+1D Minkowski space, with potential $V(\Phi) = m^2(1 - \cos 7\Phi)/49$. This level has exact Lorentz invariance ($\omega^2 = k^2 + m^2$, Lean-certified); carries the BPS kink spectrum with $\mathbb{Z}_7$ superselection; and hosts the quantum amplitudes ϕ_k from which the Born rule $P(k) = |\phi_k|^2$ follows.

The two levels are connected by the continuum limit $\varepsilon_0(M) \rightarrow 0$ as $M \rightarrow \infty$ (Theorem 5.2), the event-triggered coupling (Theorem 2.9), and the PSC/D2 Lorentz equivariance theorem (Theorem 5.1).

Two derivations of Lorentz-related physics at Level 1 are complementary, not equivalent. This paper derives **observer-level Lorentz**: PSC Presentation Invariance forces **[D]**-selected observables on the Φ_{MDL} continuum to be Lorentz-equivariant despite the Level-1 preferred frame. P36 [16] derives **dynamics-level SR**: in the Asynchronous Fractal Cellular Automaton (AFCA), each cell's inner Rule 110 clock measures a state-dependent recurrence period τ_c , and a moving glider's elevated structural complexity gives $\tau_c(\text{glider})/\tau_c(\text{ether})$ tracking γ . The CMCA (Section 3) unifies both mechanisms in a single Level-1 construction.

The Algebraic Lifting Theorem and Algebraic Descent Theorem of P28 [13] transfer propositional content over the $\mathbb{Z}_7$ orbit algebra ($\mathbb{Z}_7$ winding conservation, charge quantisation, three-generation count, color confinement, scattering vertex catalog, Garden-of-Eden stability). These theorems explicitly do not transfer geometric or dynamical content: the SR Nyquist correction $\varepsilon_0(M) = \pi^2/(3M^2)$ is M -dependent and does not descend; exact Lorentz invariance is recovered only in the $M \rightarrow \infty$ continuum limit on Φ_{MDL} . A Lean-certified theorem (`no_finite_ca_exact_lorentz_replica`, zero sorry) establishes this rigorously: $\varepsilon_0(M) = \pi^2/(3M^2) > 0$ for all $M > 0$, so no cellular automaton at any finite resolution M exactly replicates the Lorentz invariance of Φ_{MDL} ; exact invariance is recovered only as $M \rightarrow \infty$, at which point the CMCA identifies with Φ_{MDL} itself (`phimdl_is_unique_exact_lorentz_model`, zero sorry).

Explicit continuum-limit descent map. The $M \rightarrow \infty$ continuum limit is explicitly verified at $M = 7$. The Cook A-type glider winding profile (sampled over a 60-cell window centred on the glider’s centre of mass) matches the Φ_{MDL} BPS kink $\phi_{\text{kink}}(x) = (4/7) \arctan(e^{m_\phi x})$ with RMSD = 5.34%, strictly below the certified Nyquist residual $\varepsilon_0(7) = \pi^2/147 \approx 6.71\%$; the Pearson correlation between the discrete and continuum profiles is $r = 0.994$; and the winding number $Q = 1/7$ exactly, matching the $\mathbb{Z}_7$ kink charge (script `cmca_algebraic_descent.py`; results JSON `cmca_algebraic_descent_results.json`). This closes the explicit numerical bridge between the Level-1 CMCA discrete substrate and the Φ_{MDL} continuum terminal object.

10.2 Comparison with Cook (2004) and GF(7) universality

Cook (2004) [3] proved that Rule 110 is Turing-universal using a constructive Turing-complete simulation (CTS) construction. P30 [19] provides a Lean 4 formalisation of this theorem. The algebraic route via GF(7) polynomial universality [10] yields a Cook-independent Lean certificate: `z7_prime_field_universality`, `rule110_z7_poly_rep`, `rule110_center1_is_nand` (zero sorry).

In the two-layer context, Rule 110 Turing universality is the foundation for the *computational sufficiency* claim (P28 [13]): any finite-dimensional SM particle configuration can be realised as a CA excitation, because the universal computation provided by Rule 110 can simulate any computable physical process. The two-layer extension does not alter this certificate; it adds the chiral partner layer required for V–A structure and the symmetric light cone.

10.3 GTE framework context

This paper is positioned within the UGP Physics programme as follows:

- **P28** (computational universality): establishes Rule 110 as the MDL-minimal universal CA; foundational for all three CMCA layers. The Algebraic Lifting and Descent theorems certify the algebraic content-preservation morphisms in the GTE category of Level-1 CAs.
- **P30** [19] (Cook theorem certification): machine-certifies Cook’s Rule 110 universality proof; complements the GF(7) polynomial route of P40.
- **P34** (GTE-Möbius architecture): defines the architecture $(A, e, [\mathbf{D}])$ and the D2 Presentation Invariance axiom from which Theorem 5.1 derives Lorentz equivariance.
- **P35** (GTE unification): derives the Weinberg angle and EW threshold; Theorems 6.1–6.4 here provide the three-layer CMCA derivation complement.
- **P36** [16] (emergent gravity and AFCA): derives dynamics-level SR time dilation via the AFCA τ_c inner clock mechanism, complementary to the observer-level Lorentz derivation here. The CMCA (Section 3) combines both mechanisms in a single Level-1 construction.

- **P37** (quantum mechanics from Rule 110): develops the 't Hooft cogwheel construction; Born rule and double-slit results here extend P37 with the three-layer CMCA chirality structure.
- **P40** (GF(7) polynomial universality): certifies Cook-independent Turing universality; complements the Rule 124 and inner-clock construction here.
- **P43** [12] (complete Φ_{MDL} framework): proves the Algebraic Necessity Theorem — any QFT with three generations, $b_0 = 7$, and a $\mathbb{Z}_7$ -kink Born rule necessarily has $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ as its internal symmetry group (Lean-certified, `AlgebraicNecessityTheorem.lean`, zero sorry).
- **P45** [24] (three-tape CMCA): extends the 1 + 1D architecture of this paper to 3 + 1D via three MCAs sharing a common outer clock τ_c^{out} . The Dimensional Protocol Principle (DPP)—that the three tapes synchronize through this shared clock—is shown there to be both necessary and sufficient for 3 + 1D Minkowski dynamics to emerge from the three-tape system.

10.4 Open questions

Several open questions remain.

Unconditional CatAL for absolute masses. Steps 5–8 of Table 8 carry CatAD status. Achieving unconditional CatAL for M_W and M_Z in GeV requires: (1) a first-principles derivation of Δr from a CA Schwinger loop (currently CatD); (2) closure of the v_{PSC} open axiom `psc_ew_entropy_maximization` in the SRRG formalism.

Discharge of kernel-workaround axioms. The module previously carried seven private kernel-workaround axioms, all of which have been discharged via parametric proofs over `Fin n` in `PSCEffectMeasureGeneric.lean`. The sole remaining imported axiom is `re_trace_psd_mul_psd_nonneg` from `nems-lean`.

Canonical kink quantization. The canonical $\mathbb{Z}_7$ -KG kink Fock construction (Jackiw–Rebbi [6] zero-mode; MDL weight $= |\phi_k|^2$ identification in the continuum quantum limit) remains an open direction for Lean formalization in future work.

Conjecture C2 and absolute state selection. Theorem 4.2 and `CMCAMDLMinimality.lean` are conditional $\mathbf{A}_{\text{Lean}}$ (six physical-interpretation axioms, zero sorry). Full closure of P34 Conjecture C2—uniqueness of the physical $D \in [\mathbf{D}]$ —requires discharging the `architectural_restriction` axiom in `CoherenceMeasureUniqueness.lean` via a DSAC representation theorem on Φ_{MDL} .

Canonical inner-clock formalisation. The inner Rule 110 τ_c clock (Section 3) is characterised computationally (CatA). A Lean formalisation of the τ_c first-passage recurrence time and its dependence on glider vs. ether structural complexity would provide a CatAL certificate for Proposition 3.3.

Gravitational sector. The Ollivier–Ricci curvature of the Rule 110 causal graph discriminates gliders from ether ($\kappa(\text{EE}) = 0$, $\kappa(\text{SD}) = +0.769$, Gorard sign at $p = 2.5 \times 10^{-36}$, CatA [5]). Dynamical glider–glider gravitational attraction via $f(\kappa)$ edge-weight coupling was not confirmed across several distinct coupling prescriptions: both static inter-defect curvature gradients and dynamical backreaction methods yielded no measurable attraction signal. The primary gravity derivation path is the Φ_{MDL} /Einstein track (P38, in preparation).

Dimensional dissipation staircase and higher-dimensional extensions. The CMCA established in this paper is a $1 + 1\text{D}$ construction. Systematic searches have now mapped the full dimensional spectrum of $\text{GF}(7)$ -polynomial-derived rules:

- **$1 + 1\text{D}$.** The CMCA (Rule $110 \oplus$ Rule $124 \oplus$ inner τ_c) supports persistent propagating gliders indefinitely (established; **A**).
- **$2 + 1\text{D}$.** Canonical $2 + 1\text{D}$ extensions of the $\mathbb{Z}_7$ polynomial rule—axis-decomposed chiral application along x and y , a 5-layer chiral construction matching the four discrete null directions of $2 + 1\text{D}$ Minkowski plus an inner clock, and the $\mathbb{Z}_7$ von Neumann totalistic—all fail to produce stable propagating gliders. A search over four rule variants and approximately 330 random and structured initial conditions run to $T = 1,000$ finds zero persistent localised propagators. A Conway B3/S23 positive control correctly detects its canonical glider (diagonal speed $c/4$), confirming the methodology. The 5-layer construction is structurally well-defined but its layers decouple into independent $1 + 1\text{D}$ dynamics along each axis; no inter-axis coupling is derivable from the underlying polynomial. The axis-XOR rule exhibits a transient marginal regime at intermediate times that decays to vacuum at long times, qualitatively intermediate between the $1 + 1\text{D}$ edge-of-chaos regime and the pure vacuum-or-chaos bifurcation of the $3 + 1\text{D}$ extension.
- **$3 + 1\text{D}$.** The canonical 3D extension `step_fmd13d` has Langton $\lambda = 0.8575$, statistically indistinguishable within 0.05% from the random-rule expectation $6/7$ (verified exhaustively over all $7^7 = 823,543$ input configurations); the output distribution is uniform, the signature of maximum-entropy chaos saturation. A λ -scan over 510 independent outer-totalistic $\mathbb{Z}_7$ von Neumann 6-neighborhood rule-and-seed combinations finds zero Class 4 (edge-of-chaos, glider-supporting) trajectories; rules transition sharply from Class $1/2$ (ordered) at $\lambda < 0.54$ directly to Class 3 (chaos) at $\lambda > 0.54$ with no marginal window.

This constitutes a *dimensional dissipation staircase*: $\text{GF}(7)$ polynomial-derived rules support persistent gliders in $1 + 1\text{D}$, produce only transient marginal dynamics in $2 + 1\text{D}$, and exhibit only vacuum-or-chaos bifurcation in $3 + 1\text{D}$. The structural root cause is that the $\text{GF}(7)$ polynomial $p(L, C, R)$ encodes a single causal axis; in $3 + 1\text{D}$ the null cone is S^2 , which cannot be discretised without breaking Lorentz symmetry. Three converging factors account for the $3 + 1\text{D}$ vacuum-or-chaos phase: higher dimensionality (6 neighbours versus 2 in $1 + 1\text{D}$) yields a sharper percolation transition on the cubic lattice; the $\mathbb{Z}_7$ alphabet concentrates the attractor to vacuum or saturated chaos; and outer-totalistic neighbour-symmetrisation destroys chirality, which is necessary for edge-of-chaos glider transport in $1 + 1\text{D}$ rules.

The Class 4 impossibility is partially Lean-certified: `outer_totalistic_is_reflection_invariant` (zero sorry) proves that outer-totalistic rules are reflection-symmetric and therefore cannot be chiral; `no_class4_outer_totalistic_z7_3d` (**A**_{Lean} conditional; zero sorry in proof body; one named physics axiom `chirality_necessary_for_class4`) uses this to certify the absence of Class 4 behaviour in the full $\mathbb{Z}_7$ -vN6 outer-totalistic family, conditional on the named axiom that chirality is necessary for Class 4 dynamics (`Universality/NoClass4OuterTotalisticZ7`).

The framework conclusion is that no canonical $2 + 1\text{D}$ intermediate discrete substrate exists, and no canonical $3 + 1\text{D}$ discrete substrate is required or derivable from a single $\text{GF}(7)$ polynomial rule. The discrete counterpart of a $3 + 1\text{D}$ Φ_{MDL} kink pair is the $1 + 1\text{D}$ CMCA glider, transferred to $3 + 1\text{D}$ spacetime via the Algebraic Lifting Theorem. The $3 + 1\text{D}$ spectral dimension $d_s \rightarrow 4$ is Lean-certified (`spectral_dimension_limit`, **A**_{Lean}) and the algebraic continuum limit $\text{CMCA} \rightarrow \Phi_{\text{MDL}}$ is certified (`CMCAContinuumLimit.lean`, **A**_{Lean} conditional). Since Φ_{MDL} in $3 + 1\text{D}$ carries exact Poincaré symmetry and SR time dilation (`poincare_invariance_of_kg`), both follow for the $3 + 1\text{D}$ CMCA in the continuum limit by algebraic descent. A distinct and complementary route to $3 + 1\text{D}$

is established in [24]: three $1 + 1\text{D}$ CMCA's sharing a common outer clock τ_c^{out} (the Dimensional Protocol Principle) jointly produce $3 + 1\text{D}$ Minkowski dynamics, with spatial structure arising precisely from the synchronization imposed by the shared clock.

Holographic structure. Of three holographic interpretations examined—(H_1) the $1 + 1\text{D}$ CMCA as boundary of a $3 + 1\text{D}$ bulk, (H_2) a $2 + 1\text{D}$ discrete CA as boundary, and (H_3) $2 + 1\text{D}$ Φ_{MDL} continuum domain walls as boundary—only H_3 is consistent with the framework. H_1 fails on entropy count (a 1D boundary has insufficient degrees of freedom for a 3D bulk). H_2 fails because no canonical $2 + 1\text{D}$ discrete substrate exists. H_3 is consistent with established black-hole holography ($S = A/4$): the Φ_{MDL} 2D domain walls, with surface tension $\sigma = 290.10 \text{ MeV/fm}^2$, are codimension-1 surfaces in the $3 + 1\text{D}$ bulk with the appropriate entropy count to act as information-bearing boundaries. The GTE-native holographic structure is therefore continuum, not discrete.

11 Conclusion

We have presented the Chiral Minkowski Cellular Automaton (CMCA)—Rule 110 (x^+ null layer) paired with its spatial mirror Rule 124 (x^- null layer) and a shared inner Rule 110 τ_c clock (t timelike layer)—as the minimal Level-1 GTE discrete substrate simultaneously accounting for Lorentz invariance, the electroweak scale, the Born rule, V–A parity structure, and dynamics-level SR proper-time dilation.

The central conceptual point is that *Lorentz invariance is an observer effect, not a fundamental symmetry of the substrate*: the CA beable layer has a preferred frame at the Planck scale, but PSC Presentation Invariance (D2 of P34) forces all $[\mathbf{D}]$ -selected observables to be Lorentz-equivariant on the Φ_{MDL} continuum (observer-level, machine-certified in Lean 4 with zero sorry). Simultaneously, the inner τ_c clock implements proper-time dilation for moving gliders at the Level-1 dynamics (SR error 5.79%, within the $M=7$ Nyquist floor; CatA).

The quantitative results—Weinberg angle $\sin^2\theta_W = 0.231207$ (Wolfenstein $\mathbf{A}_{\text{Lean}}$; $+0.038\sigma$ at $\mathbf{B}$ with two-loop threshold [11]) from PDG 2024), $M_Z = 91.91 \text{ GeV}$ ($+0.80\%$), Born rule $P(k) = |\phi_k|^2$, double-slit $\text{corr} = 0.998$ —all follow from the orbit arithmetic of the two outer layers with no tunable free parameters.

The three-layer structure (R110 outer, R124 outer, inner R110 τ_c) corresponds exactly to the three irreducible causal directions of $1 + 1\text{D}$ Minkowski spacetime (x^+ , x^- , t). This is the Law of Threes at the level of the spacetime substrate, cascading from $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$. The CMCA is the minimal Level-1 GTE CA with all five structural features; Φ_{MDL} is its continuum terminal object in the GTE category of Level-1 constructions (Lean-certified: `GTECategoryStructure.lean`, conditional CatAL).

The CMCA is a falsifiable framework. The definitive test is the Planck-scale Lorentz violation prediction $\delta_{\text{LV}}(E) = C_\gamma(E/E_P)^2$ with $n = 2$ and $C_\gamma = \pi^2/147$, accessible to ultra-high-energy photon time-of-flight experiments at $E \gtrsim 10^{17} \text{ eV}$.

The $1 + 1\text{D}$ architecture established here is generalized in [24] to $3 + 1\text{D}$ via three CMCA's sharing a common outer clock τ_c^{out} ; the shared clock, formalized as the Dimensional Protocol Principle, is shown there to be both necessary and sufficient for $3 + 1\text{D}$ Minkowski dynamics to emerge from the three-tape system.

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A Unified CMCA Subsystem Verification

Each structural feature of the CMCA was first established in isolation (either in a single-layer construction or in a companion paper) and then verified to survive intact in the full three-layer CMCA system. Table 10 summarises these reproduction tests.

Table 10: CMCA subsystem reproduction: each property tested in isolation (prior paper or single-layer system) and in the full CMCA (UNIDM1 prototype, L=840, M=7, N_trans=300).

Property	Isolated test	CMCA test	Match?
$ v_R = v_L = 2/3$	Lean: ChiralMirror-SpeedSymmetry	UNIDM1 two-layer outer reference	✓
$\mathbb{Z}_7$ generation orbit (3-step)	GUTStructure §41	UNIDM1 $\text{gen}_1 \rightarrow \text{gen}_2 \rightarrow \text{gen}_3 \rightarrow \text{vac}$ (3 steps; mirror 2)	✓
V–A 32/125 mismatch	Lean: ChiralPairVA (zero sorry)	UNIDM1: 32 mismatches, all W^+ non-centre	✓
$\tau_c \approx \gamma$ (SR error 5.79%)	P36 [16] AFCA single-layer	UNIDM1 canonical C2 glider seed: ratio 1.563 vs $\gamma = 1.659$	✓
Sync gives no dilation	sync/async null test (§3.4)	UNIDM1 sync mode: ratio 1.000 exactly	✓
Observer-level Lorentz	Lean: PSCPILorentzMain (1 axiom)	Still applies to [D] -observables on Φ_{MDL}	✓
Turing universality	P28, P30, P40 (Rule 110, all CatAL)	Inherited by CMCA Layer 1	✓
Born rule $P(k) = \phi_k ^2$	Lean: born_rule_unconditional (zero custom physics axioms)	Carried from single-layer; $\mathbb{Z}_7$ -sector Born weights unchanged	✓
Frame duality $\text{R124}(\text{rev}(X)) = \text{rev}(\text{R110}(X))$	Computational verification (30 steps)	Structural identity of mirror rule	✓

The CMCA inherits all nine properties simultaneously. No single prior construction achieves all nine: single-layer Rule 110 (P28) lacks V–A and symmetric light cone; the two-layer system without the inner clock lacks dynamics-level SR; the AFCA of P36 lacks the symmetric light cone and V–A. The CMCA is the unique construction in the survey that passes all rows.

All computation scripts for the UNIDM1 prototype are in the paper’s **scripts/** subdirectory (see **REPRODUCE.md**). The sync/async comparison is **sync_vs_async_three_layer.py**; the single-script verification of all nine headline claims is **cmca_full_reproducibility.py**.

Wolfram Language implementation. A Wolfram Language implementation of the CMCA and all nine P41 reproducibility tests is available as **cmca_full_reproducibility_wolfram_version.wl**

in the same `scripts/` directory. It uses native `CellularAutomaton[110, ...]` and `CellularAutomaton[124, ...]` primitives for the outer chiral layers, with the inner τ_c Rule 110 clock via asynchronous table updates. All nine claims pass in approximately 41 seconds via `wolframscript -file cmca_full_reproducibility_wolfram_version.wl`.

B Lean 4 Theorem Inventory

All Lean 4 modules are in the `ugp-lean` repository. Table 11 lists all theorems used in this paper.

Table 11: Lean 4 theorem inventory for P41.

Theorem (Lean name)	Module path	Sorry	Notes
<code>rule124Fin2_is_spatial_mirror</code>	<code>Universality/</code> <code>ChiralMirrorSpeedSymmetry</code>	0	Mirror rule identity
<code>mirror_rule_speed_symmetry</code>	<code>Universality/</code> <code>ChiralMirrorSpeedSymmetry</code>	0	$ v_R = v_L = 2/3$
<code>orbit_depth_ether_period_z7_identity</code>	<code>Universality/</code> <code>OrbitDepthEtherPeriod</code>	0	depth = period = 7
<code>gcd_glider_ether</code>	<code>Universality/CouplingNoGo</code>	0	$\gcd(3, 14) = 1$
<code>orbit_length_eq_seven</code>	<code>Universality/CouplingNoGo</code>	0	period-7 orbit length
<code>gcd_glider_orbit</code>	<code>Universality/CouplingNoGo</code>	0	$\gcd(3, 7) = 1$
<code>lcm_glider_orbit</code>	<code>Universality/CouplingNoGo</code>	0	$\text{lcm}(3, 7) = 21$ sublattice obstruction
<code>coupling_no_go_cell_level</code>	<code>Universality/</code> <code>DynamicalCouplingBridge</code>	0	Cell-level no-go
<code>event_triggered_coupling_period</code>	<code>Universality/</code> <code>DynamicalCouplingBridge</code>	0	$\text{lcm}(3, 7) = 21$ resonance
<code>coupling_no_go_escape_via_event</code>	<code>Universality/</code> <code>DynamicalCouplingBridge</code>	0	Class B escape
<code>minkowski_metric_def</code>	<code>Universality/LorentzInvariance</code>	0	$\eta = \text{diag}(-1, 1, 1, 1)$
<code>lorentz_transform_preserves_inner</code>	<code>Universality/LorentzInvariance</code>	0	η invariance
<code>kg_dispersion_lorentz_invariant</code>	<code>Universality/LorentzInvariance</code>	0	On-shell $\rightarrow$ on-shell
<code>poincare_invariance_of_kg</code>	<code>Universality/LorentzInvariance</code>	0	Full Poincaré
<code>psc_pi_forces_d_lorentz_equivariant</code>	<code>Substrate/PSCPI LorentzMain</code>	0	1 axiom: <code>d2_universal</code>
<code>psc_pi_no_preferred_frame</code>	<code>Substrate/PSCPI LorentzMain</code>	0	No preferred frame
<code>orbit_absorption_at_nngen</code>	<code>Universality/GUTStructure §41</code>	0	$k = N_{\text{gen}}$ vacuum reach

(continued on next page)

(Table 11 continued)

Theorem (Lean name)	Module path	Sorry	Notes
isEWThresholdStep	Universality/GUTStructure §41	0	EW step uniqueness
ew_threshold_definitional_route	Universality/GUTStructure §41	0	Definitional $\mathbf{A}_{\text{Lean}}$
weinberg_angle_closure	Universality/GUTStructure §12	0	$\sin^2\theta_W = 3/13$
weinberg_two_term_prediction	Universality/GUTStructure §24	0	384729/1664000
mw_mz_squared_from_weinberg	Universality/GUTStructure §45	0	$(M_W/M_Z)^2 = 10/13$
e0_schwinger_sm_identity	Universality/ EWScalePrediction	0	$E_0 = v_{\text{PSC}}\sqrt{\pi/8}$
mw_formula_from_vH	Universality/ EWScalePrediction	0	M_W formula
mz_formula_from_vH	Universality/ EWScalePrediction	0	M_Z formula
ew_scale_prediction_summary	Universality/ EWScalePrediction	0	Summary bundle
beableWindingPartitionInstance	Universality/ BeableWindingPartitionInstance	0	Zero custom physics axioms
born_rule_unconditional	Universality/ BeableWindingPartitionInstance	0	Unconditional Born rule
structuralBeableAmplitude	Universality/ BeableWindingPartitionInstance	0	Structural Born
fock_beable_amplitude_normalized	Universality/ BeableWindingPartitionInstance	0	Normalization
born_rule_from_dual_frame_master	Universality/ DualFrameBornRule	0	Dual-frame Born master bundle
two_function_picture	Substrate/ TransputationStateSelector	0	Born o transputation
transputation_state_selector_bundle	Substrate/ TransputationStateSelector	0	Two-function selector ($\mathbf{A}_{\text{Lean}}$ conditional)
born_weight_from_md1_minimality	Universality/BornRuleMDL	0	Sector weights
born_rule_from_psc_md1	Universality/BornRuleMDL	0	Conditional Born
bit_mismatch_iff	Universality/ChiralPairVA	0	Mismatch criterion
va_mismatch_count	Universality/ChiralPairVA	0	Count = 32

(continued on next page)

(Table 11 continued)

Theorem (Lean name)	Module path	Sorry	Notes
pascal_row3_absorption_property	Universality/GUTStructure §37	0	Pascal/orbit joint
algebraic_descent_theorem	Universality/ AlgebraicDescentTheorem	0	F_{21} content resolution- independent
sr_accuracy_depends_on_M	Universality/ AlgebraicDescentTheorem	0	Nyquist correction M -dependent
nyquist_correction_denominator_at_m7	Universality/ AlgebraicDescentTheorem	0	$\varepsilon_0(7) = \pi^2/147$
gte_level1_category_certificate	Framework/ GTECategoryStructure	0	GTE Level-1 category; Φ_{MDL} terminal object; 4 axioms
objects_with_all_features	Framework/ GTECategoryStructure	0	Level-1 objects with all 5 features
cmca_is_md1_minimal_with_all_five_features	Framework/ CMCAMDLMinimality	0	MDL minimality master; 6 axioms
K_CA_all_features_lower_bound	Framework/ CMCAMDLMinimality	0	Lower bound = 19 bits
phimdl_thermal_state_master	Universality/ PhiMDLThermalState	0	L3 thermal state; zero custom axioms
thermal_state_sector_prob	Universality/ PhiMDLThermalState	0	$P(k) = e^{-M_k/T}/Z_T$
thermal_state_vacuum_dominant	Universality/ PhiMDLThermalState	0	Vacuum-dominant at low T
cmca_continuum_limit_is_phimdl	Framework/ CMCAContinuumLimit	0	CMCA $\rightarrow \Phi_{\text{MDL}}$ bundle; 4 inherited axioms
phimdl_lorentz_correction_tendsto_zero	Framework/ CMCAContinuumLimit	0	$\varepsilon_0(M) \rightarrow 0$ as $M \rightarrow \infty$
c2_distinguishability	Substrate/ CoherenceMeasureUniqueness	0	Six distinct $D \in [\mathbf{D}]$ candidates
c2_uniqueness_structural_bundle	Substrate/ CoherenceMeasureUniqueness	0	C2 structural reduction; 1 axiom
no_finite_ca_exact_lorentz_replica	Framework/ CMCAContinuumLimit	0	$\varepsilon_0(M) > 0$ for all $M > 0$; no finite- M CA is Lorentz-exact

(continued on next page)

(Table 11 continued)

Theorem (Lean name)	Module path	Sorry	Notes
phimdl_is_unique_exact_lorentz_model	Framework/ CMCAContinuumLimit	0	Unique zero-error Lorentz limit; bundle of the two above
outer_totalistic_is_reflection_invariant	Universality/ NoClass4OuterTotalisticZ7	0	OT rules reflection-symmetric $\Rightarrow$ not chiral
outer_totalistic_not_chiral	Universality/ NoClass4OuterTotalisticZ7	0	Corollary: OT rules cannot be chiral
outer_totalistic_z7_vn6_rule_space_card	Universality/ NoClass4OuterTotalisticZ7	0	$ \mathcal{R}_{\text{OT}} = 7^{49}$
no_class4_outer_totalistic_z7_3d	Universality/ NoClass4OuterTotalisticZ7	0	1 axiom: <code>chirality_necessary_for_class4</code>

Lean build verification. All modules compile with:

```

cd ugp-lean
lake build UgpLean.Universality.LorentzInvariance
lake build UgpLean.Substrate.PSCPILorentzMain
lake build UgpLean.Universality.EWScalePrediction
lake build UgpLean.Universality.BeableWindingPartitionInstance
lake build UgpLean.Universality.BornRuleMDL
lake build UgpLean.Universality.ChiralPairVA
lake build UgpLean.Universality.CouplingNoGo
lake build UgpLean.Universality.DynamicalCouplingBridge
lake build UgpLean.Universality.ChiralMirrorSpeedSymmetry
lake build UgpLean.Universality.AlgebraicDescentTheorem
lake build UgpLean.Framework.GTECategoryStructure
lake build UgpLean.Framework.CMCAMDLMinimality
lake build UgpLean.Framework.CMCAContinuumLimit
lake build UgpLean.Universality.PhiMDLThermalState
lake build UgpLean.Universality.DualFrameBornRule
lake build UgpLean.Substrate.TransputationStateSelector
lake build UgpLean.Substrate.CoherenceMeasureUniqueness
lake build UgpLean.Universality.NoClass4OuterTotalisticZ7

```

Sorry count: zero. Disclosed axioms: `d2_universal` (D2 of P34, PSC Presentation Invariance in universal form); `discrete_w_vertex_interpretation` (W -boson vertex as discrete inter-layer event); `architectural_restriction` (C2a Path 4c, `CoherenceMeasureUniqueness.lean`); `chirality_necessary_for_class4` (chirality necessary for Class 4 dynamics in 3D von Neumann CA; `NoClass4OuterTotalisticZ7.lean`); six physical-interpretation axioms in `CMCAMDLMinimality.lean`; four physical-interpretation axioms in `GTECategoryStructure.lean`. The Born-rule modules `BeableWindingPartitionInstance.lean`, `DualFrameBornRule.lean`, and `PhiMDLThermalState.lean` carry zero custom physics axioms. Theorems 4.3–4.5 (Algebraic Lifting,

Descent, and SR accuracy) are proved in P28 [13]; their Lean modules are `LiftingTheorem.lean` and `AlgebraicDescentTheorem.lean` in the `ugp-lean` repository.